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Price manipulation theorems for nonlinear transient impact models

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Abstract

We study price manipulation in transient market impact models with a nonlinear instantaneous impact law. We present a new theorem excluding price manipulation when slippage contains a rate-proportional concession, a fixed spread, or both. For power-law impact with exponent $1-\varepsilon \in (0,1)$, we express the exact joint slippage threshold in terms of a constant Ξ_ε, defined by optimization over round trips. For permanent square-root impact, a geometric rearrangement and a pointwise inequality yield an algebraic value for $\Xi_{1/2}$. We also show that, for square-root impact with both forms of slippage present, any criterion based on a threshold for the deviation from a single linear law can lower our sufficient bound on the product of the two slippage coefficients by at most a factor of four.

Keywords: optimal execution; transient impact; slippage; price manipulation.

JEL Classification: C61; G14.

1. Introduction

Gatheral (2010) proposed a continuous-time market impact model with nonlinear instantaneous impact and a general decay kernel. The stock price process is

$$S(t) = S(0) + \int_0^t G(t-s) f(v(s))\, ds + \int_0^t \sigma\, dW_s, \tag{1}$$

where W is Brownian motion, $v(s)$ is the rate of trading at time $s < t$ in shares per unit time, $f(v(s))$ represents the impact of trading at time s and $G(t-s)$ is a decay factor which models the transient nature of price impact; see Bouchaud (2010) for a survey and related literature.

The decay factor in (1) follows from the earlier model of Bouchaud et al. (2004), who found that the response of the price to an individual order decays slowly, roughly as a power law, and that this decay is what keeps prices diffusive in the presence of long-memory order flow. The nonlinear instantaneous law descends from the observation that impact grows as a concave function of order size, first for single trades (Lillo et al., 2003) and then for metaorders, whose impact is close to a square root over a large range of order sizes (R. Almgren et al., 2005; Tóth et al., 2011). Kyle and Obizhaeva (2016) connect this regularity to invariance principles. Bucci et al. (2019) locate the crossover from linear to square-root behaviour at small sizes, while Zarinelli et al. (2015) document departures at large participation rates. Webster (2023) provides a practitioner's synthesis.

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The theoretical literature on optimal execution has, for the most part, kept the instantaneous law linear while allowing more general decay. Obizhaeva and Wang (2013) introduced the exponentially resilient order book, and Alfonsi et al. (2010) extended the model to general book shapes. Alfonsi et al. (2012) related the existence of optimal strategies to the positive definiteness of a general decay kernel. They also introduced transaction-triggered price manipulation, which we study in Section 3.5. Gatheral et al. (2012) reduced the linear problem to a Fredholm equation of the first kind, and Alfonsi and Schied (2013) and Schied and Strehle (2021) characterized the optimal execution measures for completely monotone kernels as capacitary measures. Gatheral and Schied (2013) survey this line of work. The slippage terms of the present paper have an equally long lineage: the rate-proportional concession is the temporary impact of R. Almgren and Chriss (2001), and the fixed half-spread is the simplest proportional transaction cost. R. F. Almgren (2003) extended the execution problem to nonlinear temporary impact, including power laws, and uncertainty in execution prices that increases with the trading rate.

Nonlinear instantaneous impact with decay is comparatively recent territory. Huberman and Stanzl (2004) showed that permanent impact must be linear if quasi-arbitrage is to be excluded, and the no-dynamic-arbitrage analysis of Gatheral (2010) studied the power law $f(v) \propto \operatorname{sign}(v)\,|v|^{\delta}$ in (1), relating the impact exponent δ to the decay exponent by an inequality that the square-root law with slow decay violates. Curato et al. (2017) introduce a fixed-spread cost in the same nonlinear transient model and use numerical optimization to study the spread's effect on negative-cost strategies, and Hey et al. (2025) develop an execution theory for concave impact with decay and compare their model with order data. In a multi-asset setting, Schneider and Lillo (2019) derive no-manipulation restrictions on cross-impact and show how spread costs can offset profits from the particular round trips they examine at empirically relevant trading rates.

Predictive signals introduce an additional source of expected trading gains. Lehalle and Neuman (2019) study deterministic transient-impact schedules using the signal observed at the start, and adaptive strategies under temporary impact. Neuman and Voß (2022) derive optimal strategies that update with the signal under linear temporary impact and exponential transient impact. Abi Jaber and Neuman (2025) treat a broader class of nonnegative-definite Volterra kernels satisfying square-integrability and continuity conditions. Our price process (1) has a martingale exogenous component; the expected-cost criteria below assess gains generated by trading's own impact and slippage.

The price-manipulation theorem of Section 3 gives an explicit analytic sufficient condition by comparing the impact function with any nonnegative linear law and bounding the difference by the combined rate-proportional and spread slippage. Rate-proportional slippage alone controls the relative deviation $|f(x) - \lambda x|/(\lambda x)$, while a fixed half-spread alone controls the absolute deviation $|f(x) - \lambda x|$. For a sublinear power law, the relative deviation diverges at small rates and the absolute deviation diverges at large rates. With both forms of slippage present, the criterion applies directly to the square-root law and general admissible propagators. Corollary 1 shows that an envelope threshold, even at the best comparison slope, can lower the sufficient bound on $\eta\varsigma$ for square-root impact by at most a factor of four. Section 3.4 turns to the exact threshold. For power-law impact it is governed by a single constant Ξ_ε (Theorem 3), which is bounded from below by explicit round trips and, for permanent square-root impact, determined exactly as an algebraic number by Proposition 3. For permanent square-root impact, our sufficient condition requires $\eta\varsigma \geq 0.0625\,\lambda^2 v_*$, whereas the exact threshold is approximately $0.00164535\,\lambda^2 v_*$. Thus our sufficient condition requires a slippage product approximately 38 times larger than necessary.

For permanent square-root impact, the envelope criteria studied in Section 3.3 must require $\eta\varsigma \geq 0.015625\,\lambda^2 v_*$, by Corollary 1. This minimum requirement is approximately 9.5 times the exact threshold.

For $G(t) = t^{-1/4}$, with $\lambda = v_* = 1$ and equal slippage coefficients $\eta = \varsigma = s$, let s_* denote the smallest common coefficient that excludes manipulation. Our certified lower bound in Section 3.4 and the sufficient condition (35), using the kernel bounds in Appendix B, give

$$0.02100625 < s_* < 0.3567.$$

The corresponding bounds on the critical product s_*^2 differ by a factor of approximately 290. Determining the exact threshold within this interval remains open.

2. Setup and standing assumptions

2.1. The model

Let $T > 0$ be the trading horizon in days and let $\tau \in [0, T]$ denote clock time. Throughout the paper we use dimensionless time

$$t := \tau/T \in [0, 1]. \tag{2}$$

A *trading-rate path* $v : [0, 1] \to \mathbb{R}$ specifies at each t the instantaneous trading rate $v(t)$, in shares per unit of dimensionless time, so that its integral over a subinterval of $[0, 1]$ is a number of shares; a clock-time rate of $\nu(\tau)$ shares per day corresponds to $v(t) = T\,\nu(tT)$. In practice, some execution schedules are built for a horizon of a single trading day, with clock time quoted in days, and then $T = 1$ and the dimensionless time t coincides numerically with τ.

For a horizon of T days, let f_{day} and G_{day} denote the impact function and kernel in clock time, and let η_{day} and ς_{day} denote the slippage coefficients for rates in shares per day. Changing variables in the impact integral and in the execution price gives

$$f(x) = T f_{\text{day}}(x/T), \qquad G(u) = G_{\text{day}}(Tu), \tag{3}$$

$$\eta = \eta_{\text{day}}/T, \qquad \varsigma = \varsigma_{\text{day}}. \tag{4}$$

The reference rate becomes $v_* = T v_{*,\text{day}}$, where $v_{*,\text{day}}$ is measured in shares per day. Thus $\eta v_* = \eta_{\text{day}} v_{*,\text{day}}$. Let B_τ be Brownian motion in clock time, with volatility coefficient σ_{day} in price units per square root of a day. Then $W_t := T^{-1/2} B_{Tt}$ is Brownian motion in dimensionless time. The volatility coefficient in (1) is $\sigma = \sqrt{T}\,\sigma_{\text{day}}$, and

$$\int_0^{Tt} \sigma_{\text{day}}\, dB_\tau = \int_0^t \sigma\, dW_s.$$

These conversions preserve the execution prices and total cost of the same schedule. The kernel constants must be evaluated for $u \mapsto G_{\text{day}}(Tu)$ on each horizon.

Let the impacted midprice be given by (1). A market order executes at a price $\widetilde{S}(t)$ that includes an instantaneous concession relative to the midprice, which we call *slippage*. Let $\eta, \varsigma \geq 0$ satisfy $\eta + \varsigma > 0$, and, following R. Almgren and Chriss (2001); Schied and Strehle (2021), assume the execution price is given by

$$\widetilde{S}(t) = S(t) + \eta\, v(t) + \varsigma\, \operatorname{sign}(v(t)). \tag{5}$$

Here $\operatorname{sign}(x)$ equals 1, -1, or 0 according as x is positive, negative, or zero. The term $\eta v(t)$ charges a concession proportional to the trading rate, while ς is a fixed half-spread. We call the cases $\eta > 0, \varsigma = 0$ and $\eta = 0, \varsigma > 0$ *rate-only slippage* and *spread-only slippage*,

respectively. We formulate the results for deterministic continuous trading-rate paths. Continuity is a regularity assumption on the execution schedules. Define the one-sided past-impact operator by

$$(\mathcal{P}h)(t) := \int_0^t G(t-s)h(s)\,ds.$$

The expected implementation shortfall relative to $S(0)$ is

$$\begin{aligned} C_f^{\eta,\varsigma}[v] &:= \mathbb{E}\left[\int_0^1 v(t)\big(\widetilde{S}(t) - S(0)\big)\,dt\right] \\ &= \eta\,\|v\|_2^2 + \varsigma\,\|v\|_1 + \langle v, \mathcal{P}f(v)\rangle. \end{aligned} \tag{6}$$

Here inner products and norms are over $[0,1]$. The second equality in (6) follows from $\mathbb{E}[W_t] = 0$ and Fubini's theorem.

The sufficient conditions for nonnegative round-trip cost also apply to adapted continuous trading rates under suitable integrability assumptions. To see this, let $X(t) := \int_0^t v(s)\,ds$ be the inventory of an adapted round trip, so $X(0) = X(1) = 0$. If $\mathbb{E}\int_0^1 X(t)^2\,dt < \infty$, integration by parts gives

$$\mathbb{E}\int_0^1 v(t)W_t\,dt = -\mathbb{E}\int_0^1 X(t)\,dW_t = 0.$$

The impact and slippage inequalities apply to each realized trading-rate path. Taking expectations therefore preserves the nonnegative-cost conclusion whenever these cost terms are integrable.

The values of the trading-rate path v carry units of shares per unit of dimensionless time, while powers and logarithms act on dimensionless quantities. Throughout the paper we therefore fix a *reference rate* $v_* > 0$, a constant with the units of a trading rate whose scale is set by the market. For example, one may take the average market volume traded per horizon, which is the average daily volume (ADV) when the horizon is one trading day. We build every concrete nonlinear impact function from the dimensionless ratio v/v_*. The square-root law, for example, is a constant multiple of $\sqrt{v/v_*}$. Some standard impact functions $f(v)$, the square-root law among them, are given only for $v > 0$, so a convention is needed for sells. We impose buy-sell symmetry through oddness: Huberman and Stanzl (2004) show that no-dynamic-arbitrage requires that if market impact is permanent, then

$$f(-v) = -f(v) \ \text{ for all } v \in \mathbb{R}, \tag{7}$$

and Gatheral (2010) reports consistency of (7) with his own empirical observation and thereafter takes (7) as a standing assumption. We do the same, in the form of Definition 1, and fix the conventions of Definition 2 for the propagator.

For transient impact, (7) is a modeling assumption: buying and selling at equal absolute rates produce price changes of equal magnitude and opposite sign. The permanent-impact argument cited above motivates this symmetry within its own setting.

Definition 1. *An* instantaneous impact function *shall be a continuous function* $f : (0,\infty) \to (0,\infty)$ *such that* $\lim_{v\to 0^+} f(v) = 0$. *We extend* f *to* $\mathbb{R}$ *by* $f(0) = 0$ *and* $f(-v) = -f(v)$ *for* $v > 0$, *in keeping with* (7). *Thus a buy raises the price and a sell lowers it.*

Write the linear impact law as

$$f(v) = \lambda v, \qquad \lambda > 0, \tag{8}$$

and call λ the *impact coefficient*. The impact coefficient is in units of currency per share squared, the same units as η, and is the transient counterpart of the linear price-impact coefficient of Kyle (1985). In the linear model, execution data determine the response curve $u \mapsto \lambda G(u)$, and the normalization of G determines which point of that curve is reported as λ.

Definition 2. *A* propagator *shall be a function $G : (0, \infty) \to [0, \infty)$. Furthermore, every such propagator shall be implicitly extended to the domain $\mathbb{R} \setminus \{0\}$ by using the even extension, i.e.*

$$G(u) := G(-u) \ \text{ if } \ u < 0. \tag{9}$$

Thanks to the even extension, we write $G(t-s)$ instead of $G(|t-s|)$ when $s > t$. We allow a singularity at the origin; in particular $G(0^+)$ may be finite or $+\infty$. The impact operator $\mathcal{G}$ *is the integral operator induced by the propagator, i.e. for a suitable test function v,*

$$(\mathcal{G}v)(t) := \int_0^1 G(t-s)\, v(s)\, ds\,,$$

implicitly using the even extension (9). Since G appears here as the kernel of the integral operator $\mathcal{G}$, we also refer to G as a kernel.

The instantaneous impact function f returns the rate of price displacement created by trading, in currency per share per unit of dimensionless time, and the propagator G is a dimensionless *decay factor* evaluated at a dimensionless lag. Since a lag $t - s$ is a pure number, expressions such as the power-law kernel $G(t) = t^{-\gamma}$ are dimensionally meaningful.

For every constant $c > 0$, the pair $(cf,\ G/c)$ generates the same cost as (f, G), which we refer to as *gauge freedom*. The gauge may be fixed by normalizing the decay factor: set $G(0^+) = 1$ when $0 < G(0^+) < \infty$, or set $G(t_0) = 1$ at a reference lag $t_0 > 0$ with $G(t_0) > 0$. The lag-one normalization of the discrete propagator surveyed by Bouchaud (2010) fixes the same gauge in tick time.

2.2. Main assumptions

We use a few standard analytic assumptions on the propagator kernel for the results that follow. We collect these into the following definition.

Definition 3. *A propagator G as in Definition 2 is said to be of* positive type *if*

$$\langle u, \mathcal{G}u \rangle \ \geq \ 0 \quad \text{for all } u \in L^2([0,1]). \tag{10}$$

A propagator is said to be admissible *if it is nonconstant, nonincreasing, and continuous on $(0, \infty)$, satisfies $G \in L^1_{\mathrm{loc}}([0,\infty))$, and is of positive type.*

One can verify (10) through the Bochner–Schwartz theorem (Gel'fand & Vilenkin, 1964, Vol. 4, Chapter II). If the even extension G is a tempered distribution whose Fourier transform $\widehat{G}$ is a nonnegative tempered measure, then G is of positive type. For the power law $G(t) = t^{-\gamma}$, $0 < \gamma < 1$, the even extension $|t|^{-\gamma}$ is a tempered distribution whose Fourier transform

$$\widehat{|t|^{-\gamma}}(\omega) = C\,|\omega|^{\gamma-1}, \qquad C = 2\,\Gamma(1-\gamma)\sin(\gamma\pi/2) > 0,$$

is a nonnegative tempered measure. Here Γ is the Euler gamma function.

For every propagator $G \in L^1([0,1])$, define

$$c_G := \int_0^1 G(s)\,ds \tag{11}$$

and define the future-impact operator

$$(\mathcal{P}^* v)(t) := \int_t^1 G(s-t)v(s)\,ds, \tag{12}$$

which is the $L^2([0,1])$ adjoint of $\mathcal{P}$, so that $\mathcal{G} = \mathcal{P} + \mathcal{P}^*$ on $[0,1]$.

Two further constants of the kernel enter the estimates below: one measures the transient impact that rate-proportional slippage must absorb, the other the asymmetry between past and future impact that governs the sharper bound of Proposition 1. Write $\|\cdot\|$ for the operator norm on $L^2([0,1])$ and define

$$\nu_G := \frac{1}{2}\sup\{\langle u, \mathcal{G}u\rangle : u \in L^2([0,1]),\ \|u\|_2 = 1\}, \qquad a_G := \frac{1}{2}\|\mathcal{P} - \mathcal{P}^*\|. \tag{13}$$

Since $\mathcal{G}$ is convolution on $[0,1]$ with the even extension of G, whose mass on $[-1,1]$ is $2c_G$, Young's inequality gives $\|\mathcal{G}\| \le 2c_G$, so the quadratic form in (10) is finite for every $u \in L^2([0,1])$. If G is of positive type, then $\mathcal{G}$ is self-adjoint and positive semidefinite, so the supremum in (13) is the operator norm $\|\mathcal{G}\|$, giving $2\nu_G = \|\mathcal{G}\|$. The constant a_G is half the norm of the skew part of $\mathcal{P}$. Both ν_G and a_G are dimensionless.

Lemma 1. *Let G be a propagator of positive type with $G \in L^1([0,1])$ and $c_G > 0$. Then*

$$\int_0^1 (1-u)\,G(u)\,du \;\le\; \nu_G \;\le\; \|\mathcal{P}\| \;\le\; c_G, \qquad a_G \;\le\; \|\mathcal{P}\|.$$

In particular $\nu_G > 0$. For $G \equiv 1$ one has $\nu_G = 1/2$ and $a_G = 1/\pi$.

Proof. Young's inequality for the truncated convolution gives $\|\mathcal{P}h\|_2 \le c_G\|h\|_2$, so $\|\mathcal{P}\| \le c_G$. Since $\mathcal{G} = \mathcal{P} + \mathcal{P}^*$,

$$\langle u, \mathcal{G}u\rangle = 2\langle u, \mathcal{P}u\rangle \le 2\|\mathcal{P}\|\,\|u\|_2^2,$$

whence $\nu_G \le \|\mathcal{P}\|$, and $\|\mathcal{P} - \mathcal{P}^*\| \le 2\|\mathcal{P}\|$ gives $a_G \le \|\mathcal{P}\|$. Taking $u \equiv 1$,

$$\langle \mathbf{1}, \mathcal{G}\mathbf{1}\rangle = \int_0^1\int_0^1 G(t-s)\,ds\,dt = 2\int_0^1 (1-u)\,G(u)\,du,$$

which gives $\nu_G \ge \int_0^1 (1-u)\,G(u)\,du$; this integral is positive because $G \ge 0$ and $c_G > 0$. For $G \equiv 1$,

$$\langle u, \mathcal{G}u\rangle = (\int_0^1 u)^2 \le \|u\|_2^2$$

with equality at $u \equiv 1$, so $\nu_G = 1/2$. In the case $G \equiv 1$, the skew operator

$$(\mathcal{P} - \mathcal{P}^*)h(t) = 2\int_0^t h(s)\,ds - \int_0^1 h(s)\,ds$$

is compact and normal; differentiating the eigenvalue equation shows that its eigenfunctions are $t \mapsto e^{i(2k+1)\pi t}$, $k \in \mathbb{Z}$, with eigenvalues $-2i/((2k+1)\pi)$, so its norm is $2/\pi$ and $a_G = 1/\pi$. □

For the power-law kernel $G(t) = t^{-\gamma}$, Table 1 reports c_G, ν_G, and a_G, with ν_G and a_G computed from exact integrals of the kernel over pairs of cells. The resulting symmetric

matrix preserves the positive-type quadratic form. Appendix B bounds the error from replacing the continuous operator by this matrix and verifies the matrix computations with integer and rational arithmetic. The resulting intervals certify the three decimal places shown, using 2048 cells for $\gamma = 0.1, 0.25$ and 16384 cells for $\gamma = 0.4$. The row $\gamma = 0$ follows from the exact formulas in Lemma 1.

Table 1. Kernel constants for $G(t) = t^{-\gamma}$ on $[0, 1]$.

γ	c_G	ν_G	a_G
0	1	0.500	0.318
0.1	1.111	0.585	0.355
0.25	1.333	0.763	0.425
0.4	1.667	1.046	0.518

3. Absence of price manipulation

A round trip returns the trader to the initial position. A negative expected cost therefore represents an expected profit from trading alone (Huberman & Stanzl, 2004).

Definition 4. *A* round trip *is a trading-rate path $v \in C([0, 1])$ with $\int_0^1 v(t)\, dt = 0$. A model is said to* admit price manipulation *if some round trip has strictly negative expected cost.*

Let m be a nonnegative constant with the same units as λ and η. For $r > 0$, define the *slippage-normalized deviation* from the linear law $x \mapsto mx$ by

$$\Theta_m(r) := \sup_{0<x\le r} \frac{|f(x) - mx|}{\eta x/\nu_G + \varsigma/c_G}, \tag{14}$$

with ν_G and c_G from (13) and (11). The constants ν_G and c_G weight the two forms of slippage by the size of the transient impact that each form of slippage has to absorb. The numerator and denominator in (14) both have the units of a price, so $\Theta_m(r)$ is dimensionless. Set $\Theta_m(0) := 0$, write $\Theta_{m,v} := \Theta_m(\|v\|_\infty)$, and define

$$\Theta_m^* := \sup_{x>0} \frac{|f(x) - mx|}{\eta x/\nu_G + \varsigma/c_G}. \tag{15}$$

Different choices of m give different bounds for the same cost. The condition $\Theta_m^* \le 1$ in the theorem below says that f stays within a band around the line mx whose half-width at rate x is the slippage $\eta x/\nu_G + \varsigma/c_G$ paid at that rate, reweighted by the kernel.

Theorem 1. *Let G be a propagator of positive type with $G \in L^1([0, 1])$ and $c_G > 0$, let $\eta, \varsigma \ge 0$ satisfy $\eta + \varsigma > 0$, and let f be an instantaneous impact function. For every $m \ge 0$ and every trading-rate path $v \in C([0, 1])$,*

$$C_f^{\eta,\varsigma}[v] \ge (1 - \Theta_{m,v})\,(\eta\, \|v\|_2^2 + \varsigma\, \|v\|_1). \tag{16}$$

If $\Theta_m^* \le 1$ for some $m \ge 0$, then every round trip has nonnegative expected cost and the model admits no price manipulation. If $\Theta_m^* < 1$, then every nonzero v has strictly positive expected cost.

Proof. The comparison with mx splits the impact into a nonnegative linear part and a remainder. Define

$$w(t) := f\big(v(t)\big) - mv(t).$$

Then

$$C_f^{\eta,\varsigma}[v] = \eta\, \|v\|_2^2 + \varsigma\, \|v\|_1 + m\langle v, \mathcal{P}v\rangle + \langle v, \mathcal{P}w\rangle. \tag{17}$$

Since $\mathcal{G} = \mathcal{P} + \mathcal{P}^*$,

$$\langle v, \mathcal{P}v \rangle = \frac{1}{2}\langle v, \mathcal{G}v \rangle \geq 0$$

by the positive type assumption. We next bound the remainder $\langle v, \mathcal{P}w \rangle$ in (17).

The definition of $\Theta_{m,v}$, the odd extension of f, and $f(0) = 0$ give the pointwise estimate

$$|w(t)| \leq \Theta_{m,v}\left(\frac{\eta}{\nu_G}\,|v(t)| + \frac{\varsigma}{c_G}\right), \qquad t \in [0,1]. \tag{18}$$

Kernel nonnegativity gives $|\mathcal{P}w| \leq \mathcal{P}|w|$ pointwise. The triangle inequality and (18) therefore give

$$\begin{aligned} \left|\langle v, \mathcal{P}w \rangle\right| &\leq \langle |v|, \mathcal{P}|w| \rangle \\ &\leq \Theta_{m,v}\,\frac{\eta}{\nu_G}\langle |v|, \mathcal{P}|v| \rangle + \Theta_{m,v}\,\frac{\varsigma}{c_G}\langle |v|, \mathcal{P}\mathbf{1} \rangle. \end{aligned} \tag{19}$$

Since $\mathcal{G} = \mathcal{P} + \mathcal{P}^*$, the definition (13) gives

$$\langle |v|, \mathcal{P}|v| \rangle = \frac{1}{2}\langle |v|, \mathcal{G}|v| \rangle \leq \nu_G \|v\|_2^2.$$

Also,

$$0 \leq (\mathcal{P}\mathbf{1})(t) = \int_0^t G(t-s)\,ds \leq c_G,$$

and hence

$$\langle |v|, \mathcal{P}\mathbf{1} \rangle \leq c_G \|v\|_1.$$

Substitution in (19) yields

$$\left|\langle v, \mathcal{P}w \rangle\right| \leq \Theta_{m,v}\left(\eta\,\|v\|_2^2 + \varsigma\,\|v\|_1\right).$$

Hence

$$\langle v, \mathcal{P}w \rangle \geq -\Theta_{m,v}\left(\eta\,\|v\|_2^2 + \varsigma\,\|v\|_1\right).$$

Substituting this lower bound and $m\langle v, \mathcal{P}v \rangle \geq 0$ into (17) proves (16). The two statements about Θ_m^* follow from (16) and $\Theta_{m,v} \leq \Theta_m^*$. □

For $\varsigma = 0$ and $\eta > 0$, choose $m = \lambda$ and define the relative nonlinearity up to rate r by

$$\kappa(r) := \sup_{0 < x \leq r} \frac{|f(x) - \lambda x|}{\lambda x}. \tag{20}$$

Geometrically, $\kappa(r)$ measures the fractional departure of the chord slope $f(x)/x$ from λ. Write $\kappa_v := \kappa(\|v\|_\infty)$, with $\kappa(0) := 0$. Then

$$\Theta_{\lambda,v} = \frac{\nu_G \lambda}{\eta}\kappa_v,$$

and (16) gives

$$C_f^{\eta,0}[v] \geq \left(\eta - \lambda\nu_G\kappa_v\right)\|v\|_2^2. \tag{21}$$

Thus every nonzero trading-rate path with $\kappa_v < \eta/(\lambda\nu_G)$ has strictly positive expected cost. In particular, the global condition

$$\kappa^* := \sup_{x > 0} \frac{|f(x) - \lambda x|}{\lambda x} \leq \frac{\eta}{\lambda\nu_G} \tag{22}$$

rules out price manipulation.

The proof of Theorem 1 bounds the remainder $\langle v, \mathcal{P}w\rangle$ by its absolute value. Retaining the nonnegative term $\lambda\langle v, \mathcal{P}v\rangle$ and splitting $\mathcal{P}$ into symmetric and skew parts gives a cost bound quadratic in κ_v. The coefficient of κ_v in the resulting penalty is λa_G.

Proposition 1. *Let G be a propagator of positive type with $G \in L^1([0,1])$ and $c_G > 0$, let $\eta > 0$, and let f be an instantaneous impact function. For every trading-rate path $v \in C([0,1])$,*

$$C_f^{\eta,0}[v] \geq \Big(\eta - \lambda\Big(a_G\kappa_v + \frac{\nu_G}{4}\kappa_v^2\Big)\Big)\|v\|_2^2. \tag{23}$$

Consequently the global condition

$$a_G\kappa^* + \frac{\nu_G}{4}(\kappa^*)^2 \leq \frac{\eta}{\lambda} \tag{24}$$

rules out price manipulation, and strict inequality in (24) gives strictly positive expected cost to every nonzero trading-rate path.

Proof. Write $w := f(v) - \lambda v$. By oddness of f and the definition of κ_v, $|w(t)| \leq \lambda\kappa_v|v(t)|$ for every t, so $\|w\|_2 \leq \lambda\kappa_v\|v\|_2$. Since $\mathcal{P} = \frac{1}{2}\mathcal{G} + \frac{1}{2}(\mathcal{P} - \mathcal{P}^*)$ and $\langle v, (\mathcal{P} - \mathcal{P}^*)v\rangle = 0$,

$$\langle v, \mathcal{P}f(v)\rangle = \frac{\lambda}{2}\langle v, \mathcal{G}v\rangle + \frac{1}{2}\langle v, \mathcal{G}w\rangle + \frac{1}{2}\langle v, (\mathcal{P} - \mathcal{P}^*)w\rangle. \tag{25}$$

The form $(u,h) \mapsto \langle u, \mathcal{G}h\rangle$ is symmetric and positive semidefinite by the positive type assumption, so the Cauchy–Schwarz inequality gives $|\langle v, \mathcal{G}w\rangle| \leq XY$ with $X := \langle v, \mathcal{G}v\rangle^{1/2}$ and $Y := \langle w, \mathcal{G}w\rangle^{1/2}$. Hence the sum of the first two terms in (25) is at least

$$\frac{\lambda}{2}X^2 - \frac{1}{2}XY \;\geq\; \min_{X\geq 0}\Big(\frac{\lambda}{2}X^2 - \frac{1}{2}XY\Big) = -\frac{Y^2}{8\lambda} \;\geq\; -\frac{\nu_G}{4\lambda}\|w\|_2^2,$$

where the last step uses $Y^2 \leq 2\nu_G\|w\|_2^2$ from (13). The third term in (25) is at least $-a_G\|v\|_2\|w\|_2$ by the definition of a_G. Substituting $\|w\|_2 \leq \lambda\kappa_v\|v\|_2$ into both estimates and adding $\eta\|v\|_2^2$ proves (23). The global condition (24) and the strict positivity assertion follow from $\kappa_v \leq \kappa^*$. □

Remark 1. *For $0 < \kappa^* < \infty$, condition (24) requires at most as large a value of η as (22) exactly when $\kappa^* \leq 4(1 - a_G/\nu_G)$. At $\kappa^* = 0$, both conditions reduce to $\eta \geq 0$.*

For spread-only slippage, set $\eta = 0$ in (5). Choose $m = \lambda$, and define the absolute nonlinearity up to rate r by

$$\chi(r) := \sup_{0<x\leq r}|f(x) - \lambda x|. \tag{26}$$

Write $\chi_v := \chi(\|v\|_\infty)$, with $\chi(0) := 0$. Since $\Theta_{\lambda,v} = c_G\chi_v/\varsigma$ when $\eta = 0$, (16) gives

$$C_f^{0,\varsigma}[v] \geq (\varsigma - c_G\chi_v)\|v\|_1. \tag{27}$$

Thus every nonzero trading-rate path with $\chi_v < \varsigma/c_G$ has strictly positive expected cost. The global condition

$$\chi^* := \sup_{x>0}|f(x) - \lambda x| \leq \frac{\varsigma}{c_G} \tag{28}$$

rules out price manipulation.

3.1. Power-law impact

Consider the one-parameter impact law

$$f(x;\varepsilon) := \lambda x(x/v_*)^{-\varepsilon}, \qquad x > 0, \tag{29}$$

where $\varepsilon \in \mathbb{R}$. The impact-function condition at the origin restricts $\varepsilon < 1$. For this family, the global deviations κ^* and χ^* are zero when $\varepsilon = 0$ and infinite otherwise. In the sublinear case $0 < \varepsilon < 1$, the relative deviation diverges at zero and the absolute deviation from λx diverges at infinity. This motivates using both forms of slippage: the half-spread controls the deviation at small rates, and the rate-proportional concession controls it at large rates. Section 3.4 also establishes when either form of slippage alone permits profitable round trips.

Suppose $0 < \varepsilon < 1$ and $\eta, \varsigma > 0$. Introduce the dimensionless slippage ratios

$$r_\eta := \frac{\eta}{\nu_G \lambda}, \qquad r_\varsigma := \frac{\varsigma}{c_G \lambda v_*}, \qquad A_\varepsilon := \varepsilon^\varepsilon \left(\frac{1-\varepsilon}{2}\right)^{1-\varepsilon}.$$

The ratio r_η compares the rate-proportional slippage coefficient with the spectrally weighted transient-impact coefficient $\nu_G \lambda$, while r_ς compares the half-spread with the reference-rate impact $c_G \lambda v_*$. Both ratios are invariant under the gauge freedom described in Section 2.1.

Set $z := c_G \eta x/(\nu_G \varsigma)$ and $p := (m/\lambda)(r_\varsigma/r_\eta)^\varepsilon$. Then

$$\Theta^*_m = \frac{1}{r_\eta^{1-\varepsilon} r_\varsigma^\varepsilon} \sup_{z>0} \frac{|z^{1-\varepsilon} - pz|}{1+z}.$$

The last supremum is at least p, by taking $z \to \infty$. It equals p precisely when $z^{1-\varepsilon} \le p(1+2z)$ for every $z > 0$. The maximum of $z^{1-\varepsilon}/(1+2z)$ is A_ε, attained at $z = (1-\varepsilon)/(2\varepsilon)$. Thus $p = A_\varepsilon$ gives a supremum of A_ε; for $p < A_\varepsilon$, evaluation at this same z gives a value greater than A_ε. Hence the minimizing slope is

$$\frac{m^*_\varepsilon}{\lambda} = A_\varepsilon \left(\frac{r_\eta}{r_\varsigma}\right)^\varepsilon. \tag{30}$$

The global deviation at the slope m^*_ε is

$$\inf_{m \ge 0} \Theta^*_m = \Theta^*_{m^*_\varepsilon} = \frac{A_\varepsilon}{r_\eta^{1-\varepsilon} r_\varsigma^\varepsilon}. \tag{31}$$

Consequently, Theorem 1 rules out price manipulation when

$$r_\eta^{1-\varepsilon} r_\varsigma^\varepsilon \ge A_\varepsilon. \tag{32}$$

The left-hand side of (32) is the weighted geometric mean of the rate-proportional slippage ratio r_η and the fixed-spread ratio r_ς. Both sides of (32) are dimensionless.

The square-root law is the case $\varepsilon = 1/2$ of (29). In this case the deviation from mx has an explicit expression at every comparison slope. For $\alpha, \beta > 0$ and $m \ge 0$, set

$$a := \lambda\sqrt{v_*}, \qquad y := \sqrt{\frac{\alpha x}{\beta}}, \qquad p := \frac{m\sqrt{\beta}}{a\sqrt{\alpha}}.$$

To evaluate the supremum, write $h_p(y) := (y - py^2)/(1+y^2)$. Its derivative is

$$h_p'(y) = \frac{1 - 2py - y^2}{(1+y^2)^2}.$$

Thus its maximum occurs at $y = \sqrt{1+p^2} - p$ and equals $(\sqrt{1+p^2} - p)/2$. On the region where $h_p(y) < 0$, one has $-h_p(y) < p$, with $-h_p(y) \to p$ as $y \to \infty$. Combining the two regions gives

$$\sup_{x>0} \frac{|f(x;1/2) - mx|}{\alpha x + \beta} = \frac{a}{\sqrt{\alpha\beta}} \sup_{y>0} \frac{|y - py^2|}{1+y^2} = \frac{a}{\sqrt{\alpha\beta}} \max\left\{p, \frac{\sqrt{1+p^2} - p}{2}\right\}. \tag{33}$$

The first entry in the maximum increases with p, while the second decreases. Their equality gives $3p = \sqrt{1+p^2}$. Thus the right-hand side of (33) is minimized over $p \geq 0$ at $p = 1/(2\sqrt{2})$, with minimum value $a/(2\sqrt{2\alpha\beta})$. With $\alpha = \eta/\nu_G$ and $\beta = \varsigma/c_G$ the left-hand side of (33) is Θ^*_m, the minimizing slope is

$$m_* := \frac{\lambda}{2\sqrt{2}} \sqrt{\frac{c_G \eta v_*}{\nu_G \varsigma}},$$

and

$$\inf_{m \geq 0} \Theta^*_m = \Theta^*_{m_*} = \frac{\lambda \sqrt{\nu_G c_G v_*}}{2\sqrt{2\eta\varsigma}}. \tag{34}$$

Therefore

$$8\eta\varsigma \geq \lambda^2 v_* \nu_G c_G \tag{35}$$

is sufficient to rule out price manipulation under square-root impact, for a general admissible kernel. At the boundary case $\Theta^*_{m_*} = 1$, the square-root law lies inside the band $m_* x \pm (\eta x/\nu_G + \varsigma/c_G)$. Near zero, the band width is governed by the half-spread; at large rates, it grows in proportion to the rate. The square-root law touches the upper edge of the band at $x = \nu_G\varsigma/(2c_G\eta)$. How far (35) lies from the exact threshold is taken up in Section 3.4.

3.2. Interpretation in market units

We express the sufficient condition (35) in relative-price units and execution-cost units commonly used in practice. Take a one-day horizon and choose v_* to be average daily share volume, abbreviated ADV. Let s_{BA} be the observed full bid–ask spread as a fraction of the midprice. Then

$$\frac{\varsigma}{S(0)} = \frac{s_{\mathrm{BA}}}{2}.$$

For a constant buy rate $x > 0$, the model gives the end-of-horizon impact

$$\int_0^1 G(1-s) f(x;1/2)\, ds = c_G \lambda v_* \sqrt{\frac{x}{v_*}}.$$

The empirical square-root law writes the corresponding peak impact $I_{\mathrm{peak}}(x)$ as

$$\frac{I_{\mathrm{peak}}(x)}{S(0)} = Y_0 \sigma_{\mathrm{d}} \sqrt{\frac{x}{v_*}},$$

where σ_{d} is daily volatility in relative-price units and Y_0 is the empirical impact coefficient (Bucci et al., 2019). Matching end-of-horizon impact to this law is a calibration assumption. Here Y_0 must describe constant-rate orders executed over one day, with impact measured at completion relative to the starting midprice. This matching gives

$$\frac{c_G \lambda v_*}{S(0)} = Y_0 \sigma_{\mathrm{d}}. \tag{36}$$

For another horizon, use (3) and (4) and estimate the impact coefficient for that horizon and execution profile. If the transformed kernel is renormalized by multiplying it by a positive constant, divide the transformed impact function by the same constant, as in Section 2.1. It is convenient to report the coefficient η in practitioner units. Let b_η be the number of basis points of rate-proportional execution cost for each 1% of average daily share volume traded over the day. Then

$$b_\eta := 100\frac{\eta v_*}{S(0)}. \tag{37}$$

Indeed, for every nonzero trade, (5) becomes

$$\operatorname{sign}\big(v(t)\big)\frac{\widetilde{S}(t) - S(t)}{S(0)} = \frac{s_{\mathrm{BA}}}{2} + \frac{b_\eta}{100}\frac{|v(t)|}{v_*}.$$

When the normalized trading rate $|v(t)|/v_*$ is 0.01, the rate-proportional term $(b_\eta/100)|v(t)|/v_*$ equals $b_\eta \times 10^{-4}$, or b_η basis points.

To estimate b_η as defined in (37), measure each trade's execution price relative to the midprice at execution time. Multiply this difference by the trade's sign, subtract the half-spread, and express the result in basis points relative to $S(0)$. Equation (5) gives

$$10^4\frac{\operatorname{sign}(v(t))\big[\widetilde{S}(t) - S(t)\big] - \varsigma}{S(0)} = b_\eta\left(100\frac{|v(t)|}{v_*}\right).$$

Regress this concession on the absolute trading rate expressed in percentage points of v_*, fitting a line through the origin. The slope estimates b_η. The functions f and G describe changes in the midprice through (1); that midprice is observed separately for this calculation.

Using (36) in (35) gives the following sufficient condition under this calibration assumption:

$$b_\eta \geq \frac{\nu_G}{c_G}\,\frac{25\Upsilon_0^2\sigma_{\mathrm{d}}^2}{s_{\mathrm{BA}}}. \tag{38}$$

The number b_η measures the execution-price cost ηv alone; the fixed half-spread and the transient midprice impact generated by f and G are accounted for separately. The factor ν_G/c_G lies in $(0, 1]$ by Lemma 1. Figure 1 illustrates this sufficient condition for assumed parameters. Panel (a) uses the worst-case ratio $\nu_G/c_G = 1$. Panel (b) uses $G(t) = t^{-1/4}$. For $G(t) = t^{-\gamma}$, Lemma 1 and a bound by the largest row integral give

$$\frac{1}{2-\gamma} \leq \frac{\nu_G}{c_G} \leq 2^{\gamma-1}. \tag{39}$$

Indeed, $2|u(t)u(s)| \leq |u(t)|^2 + |u(s)|^2$ bounds the quadratic form of $\mathcal{G}$ by its largest row integral times $\|u\|_2^2$, and

$$\max_{0\leq t\leq 1}\int_0^1 |t-s|^{-\gamma}\,ds = \frac{2^\gamma}{1-\gamma}.$$

For $\gamma = 1/4$, therefore, $c_G = 4/3$, $16/21 \leq \nu_G \leq 2^{5/4}/3$, and $4/7 \leq \nu_G/c_G \leq 2^{-3/4}$. Panel (b) propagates this analytic interval into the slippage bound; its upper edge is a guaranteed sufficient bound. The numerical ratio of approximately 0.5725 is shown separately. All market parameters in the figure are assumed values; the band describes uncertainty in the kernel constant. More generally, if the ratio, impact coefficient, volatility, and spread lie in positive intervals, an upper bound in (38) follows by using the upper endpoints for ν_G/c_G, Υ_0, and σ_{d}, and the lower endpoint for s_{BA}.

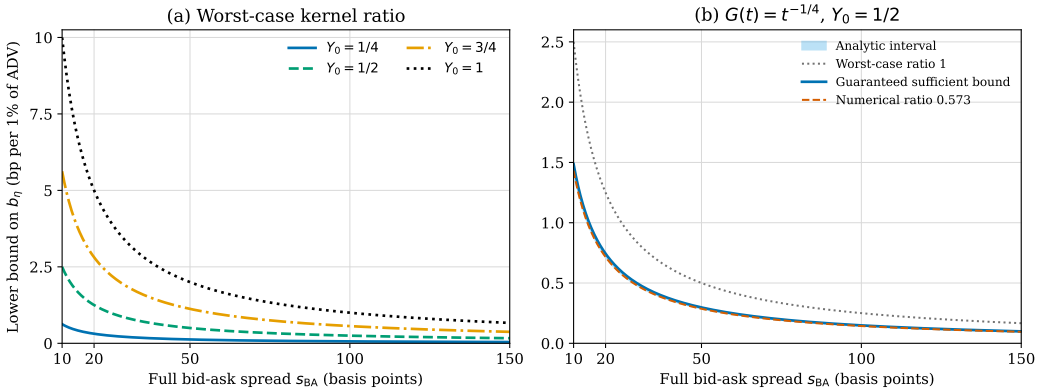


Figure 1. Sufficient lower bound on b_η from (38), $\sigma_d = 2\%$. (a) Worst-case ratio $\nu_G/c_G = 1$. (b) $G(t) = t^{-1/4}$, $Y_0 = 1/2$: analytic band from (39) and numerical estimate.

3.3. Limits of envelope criteria

Assume $\eta, \varsigma > 0$. We now ask how much the sufficient condition $\Theta_m^* \le 1$ can be relaxed when the impact function is described by the same envelope,

$$|f(x) - mx| \le \Theta_m^*(f)\,(\eta x/\nu_G + \varsigma/c_G).$$

Call a number $\theta_0 > 0$, depending on G, η, ς, and m, an *envelope threshold at slope m* if every instantaneous impact function f with $\Theta_m^*(f) \le \theta_0$ admits no price manipulation. Theorem 1 says that $\theta_0 = 1$ is an envelope threshold at every slope. Theorem 2 bounds every envelope threshold by 2. For square-root impact with $\eta, \varsigma > 0$, Corollary 1 translates this ceiling into a maximum factor-of-four reduction in the sufficient bound on $\eta\varsigma$. The ceiling applies to criteria using the scalar $\Theta_m^*(f)$ at a single slope m. Criteria using the full function $x \mapsto |f(x) - mx|$, or deviations from several slopes jointly, use additional information. The counterexample impact functions in Theorem 2 are nondecreasing and concave, so the ceiling also applies when these two properties are required. The upper bound on envelope thresholds is expressed using

$$\mu_G := \sup\left\{ \frac{-\langle v, \mathcal{P}\,\mathrm{sign}(v)\rangle}{\|v\|_1} \;:\; v \in C([0,1]) \setminus \{0\},\ \int_0^1 v(t)\,dt = 0 \right\}, \tag{40}$$

where $\mathrm{sign}(v)$ is applied pointwise. The constant μ_G is dimensionless; the numerator in (40) is the profit of the round trip v under an impact of one unit per unit time in the direction of each trade, whatever its rate.

Lemma 2. *Let G be a propagator of positive type with $G \in L^1([0,1])$ and $c_G > 0$. Then*

$$c_G/2 \le \mu_G \le c_G.$$

For $G \equiv 1$, $\mu_G = 1/2$.

Proof. Since $G \ge 0$ and $|\mathrm{sign}(v)| \le 1$, $|\mathcal{P}\,\mathrm{sign}(v)| \le \mathcal{P}\mathbf{1} \le c_G$ pointwise, so

$$-\langle v, \mathcal{P}\,\mathrm{sign}(v)\rangle \le c_G\|v\|_1$$

for every round trip, which is $\mu_G \le c_G$.

For $\mu_G \geq c_G/2$, write $\Gamma(x) := \int_0^x G(u)\,du$, and let $\Lambda_{p,\ell}$ denote the unit-mass tent function on $[p, p+\ell]$,

$$\Lambda_{p,\ell}(t) := \frac{2}{\ell}\max\Big\{0,\ 1 - \Big|\frac{2(t-p)}{\ell} - 1\Big|\Big\},$$

which is continuous, vanishes off $(p, p+\ell)$, and integrates to one. Fix $0 < 2h < \tau < 1-h$ and $\zeta > 0$, let $\phi(t) := \max\{0, \min\{t/h,\ 1,\ (\tau - t)/h\}\}$ be the trapezoid of height one on $[h, \tau - h]$ supported in $[0, \tau]$, put $q := 1 + \zeta(\tau - h)$, and define

$$v(t) := -\Lambda_{0,h}(t) - \zeta\,\phi(t) + q\,\Lambda_{1-h,h}(t), \qquad t \in [0,1].$$

Then v is continuous, $\int_0^\tau v = -1 - \zeta(\tau - h) = -q$ and $\int_{1-h}^1 v = q$, so v is a round trip with $\|v\|_1 = 2q$. The path sells one unit at once, keeps selling at the small rate ζ until time τ, and buys everything back at the end. Its sign $s := \operatorname{sign}(v)$ equals -1 on $(0,\tau)$, 0 on $[\tau, 1-h]$, and $+1$ on $(1-h, 1)$. For $t \in (0,\tau)$, $(\mathcal{P}s)(t) = -\Gamma(t)$; for $t \in (1-h, 1)$,

$$(\mathcal{P}s)(t) = -\big(\Gamma(t) - \Gamma(t-\tau)\big) + \Gamma(t-1+h) \ \leq\ -\big(\Gamma(t) - \Gamma(t-\tau)\big) + \Gamma(h).$$

Since Γ is nondecreasing with $\Gamma \leq c_G$ on $[0,1]$,

$$\begin{aligned} -\langle v, \mathcal{P}s\rangle &= -\int_0^\tau (\Lambda_{0,h} + \zeta\phi)\Gamma\,dt - q\int_{1-h}^1 \Lambda_{1-h,h}\,(\mathcal{P}s)\,dt \\ &\geq -\Gamma(h) - \zeta c_G \tau + q\Big(\min_{[1-h,1]}\big(\Gamma(t) - \Gamma(t-\tau)\big) - \Gamma(h)\Big). \end{aligned}$$

As $h \downarrow 0$, $\Gamma(h) \to 0$ and $\min_{[1-h,1]}(\Gamma(t) - \Gamma(t-\tau))$ tends to $\Gamma(1) - \Gamma(1-\tau)$ by continuity of Γ; as $\zeta \downarrow 0$, $q \to 1$; and as $\tau \uparrow 1$, $\Gamma(1) - \Gamma(1-\tau) \to c_G$. Since $\|v\|_1 = 2q \to 2$ as $h \downarrow 0$, then $\zeta \downarrow 0$, then $\tau \uparrow 1$, the ratio $-\langle v, \mathcal{P}s\rangle / \|v\|_1$ has limiting lower bound $c_G/2$. Hence $\mu_G \geq c_G/2$.

For $G \equiv 1$, $c_G = 1$, so $\mu_G \geq 1/2$. For the reverse inequality, fix a round trip v and put

$$H(t) := (\mathcal{P}\operatorname{sign}(v))(t) = \int_0^t \operatorname{sign}(v(u))\,du.$$

Since $|\operatorname{sign}(v)| \leq 1$, the function H is 1-Lipschitz, so $\max H - \min H \leq 1$. Put $a := (\max H + \min H)/2$, so that $|H - a| \leq 1/2$ on $[0,1]$. Since $\int_0^1 v = 0$,

$$-\langle v, \mathcal{P}\operatorname{sign}(v)\rangle = -\int_0^1 v(t)\big(H(t) - a\big)\,dt \leq \frac{\|v\|_1}{2}.$$

Hence $\mu_G \leq 1/2$, and so $\mu_G = 1/2$. □

Theorem 2. *Let G be a propagator of positive type with $G \in L^1([0,1])$ and $c_G > 0$, let $\eta, \varsigma > 0$, let $m \geq 0$, and let $\theta > 0$ satisfy $\theta\mu_G > c_G$. Then there is a nondecreasing concave instantaneous impact function $\tilde{f}$ with $\Theta^*_m(\tilde{f}) \leq \theta$ for which the model admits price manipulation. Consequently every envelope threshold at slope m is at most c_G/μ_G, and by Lemma 2 at most 2.*

Proof. By (40) and $\theta\mu_G > c_G$, there is a round trip $v \in C([0,1])$ with

$$-\theta\langle v, \mathcal{P}\operatorname{sign}(v)\rangle > c_G\|v\|_1. \tag{41}$$

Write $\alpha := \eta/\nu_G$ and $\beta := \varsigma/c_G$. For $\delta > 0$ let $g_\delta(x) := \min\{x/\delta, 1\}$ for $x > 0$, extended to an odd function on $\mathbb{R}$, and define

$$\tilde{f}(x) := \theta\beta\, g_\delta(x) + \max\{m - \theta\alpha, 0\}\, x, \qquad x > 0,$$

extended as in Definition 1. The function $\tilde{f}$ is continuous, positive, nondecreasing, and concave on $(0, \infty)$ with $\tilde{f}(0^+) = 0$, so $\tilde{f}$ is an instantaneous impact function. For every rate $x \geq \delta$, $\tilde{f}(x)$ is at least the fixed amount $\theta\beta$. If $m \leq \theta\alpha$, then $\tilde{f}(x) - mx = \theta\beta g_\delta(x) - mx$ lies between $-\theta\alpha x$ and $\theta\beta$; if $m > \theta\alpha$, then $\tilde{f}(x) - mx = \theta\beta g_\delta(x) - \theta\alpha x$ lies between $-\theta\alpha x$ and $\theta\beta$. In both cases $|\tilde{f}(x) - mx| \leq \theta(\alpha x + \beta)$, so $\Theta^*_m(\tilde{f}) \leq \theta$.

For $c > 0$, the path cv is a round trip, and by (6) and oddness of g_δ,

$$C^{\eta,\varsigma}_{\tilde{f}}[cv] = c^2\eta\|v\|_2^2 + c\varsigma\|v\|_1 + \max\{m - \theta\alpha, 0\}\,\frac{c^2}{2}\langle v, \mathcal{G}v\rangle + \theta\beta\, c\,\langle v, \mathcal{P}g_\delta(cv)\rangle.$$

Hence

$$\frac{C^{\eta,\varsigma}_{\tilde{f}}[cv]}{c} = c\left(\eta\|v\|_2^2 + \max\{m - \theta\alpha, 0\}\,\frac{1}{2}\langle v, \mathcal{G}v\rangle\right) + \varsigma\|v\|_1 + \theta\beta\langle v, \mathcal{P}g_\delta(cv)\rangle. \tag{42}$$

Set

$$\rho := \beta\Big(-\theta\langle v, \mathcal{P}\,\mathrm{sign}(v)\rangle - c_G\|v\|_1\Big),$$

which is positive by (41), and fix $c > 0$ so small that the first term on the right of (42) is below $\rho/2$. As $\delta \downarrow 0$, $g_\delta(cv(u)) \to \mathrm{sign}(v(u))$ for every u, with $|g_\delta| \leq 1$; by dominated convergence, first in u against $G(t - \cdot) \in L^1$ and then in t against $v \in L^1$ with the bound $|\mathcal{P}g_\delta(cv)| \leq c_G$,

$$\langle v, \mathcal{P}g_\delta(cv)\rangle \longrightarrow \langle v, \mathcal{P}\,\mathrm{sign}(v)\rangle \qquad \text{as } \delta \downarrow 0.$$

Choose δ so that $\theta\beta\langle v, \mathcal{P}g_\delta(cv)\rangle$ is within $\rho/2$ of $\theta\beta\langle v, \mathcal{P}\,\mathrm{sign}(v)\rangle = -\varsigma\|v\|_1 - \rho$. Then the right-hand side of (42) is below

$$\frac{\rho}{2} + \varsigma\|v\|_1 - \Big(\varsigma\|v\|_1 + \rho\Big) + \frac{\rho}{2} = 0,$$

so the round trip cv has strictly negative expected cost under $\tilde{f}$. An envelope threshold $\theta_0 > c_G/\mu_G$ would incorrectly rule out manipulation for the function $\tilde{f}$ constructed with $\theta = \theta_0$. Therefore $\theta_0 \leq c_G/\mu_G \leq 2$, as claimed. □

For $G \equiv 1$, Lemma 2 gives $c_G/\mu_G = 2$, so the kernel-dependent ceiling c_G/μ_G coincides with the bound 2 for permanent impact.

The counterexample in Theorem 2 works because the bound $\Theta^*_m(\tilde{f}) \leq \theta$ permits substantial impact at arbitrarily small trading rates. The constructed function satisfies $\tilde{f}(x) \geq \theta\varsigma/c_G$ for every $x \geq \delta$, however small δ is. The strategy in Lemma 2 illustrates how to exploit the impact at small rates: sell to establish a short position, continue selling slowly to depress the price, then buy back the position. By choosing δ small enough, the slow sales generate substantial impact while their spread and quadratic execution costs remain small. For square-root impact, by comparison, $f(x) = \lambda\sqrt{v_* x}$ tends to zero at a specified rate as $x \downarrow 0$. The envelope bound also admits the functions $\tilde{f}$, so a criterion based solely on Θ^*_m must account for profitable round trips under $\tilde{f}$. Figure 2(a) shows the square-root law within its optimal envelope at $\Theta^*_{m_*} = 1$. Figure 2(b) shows $\tilde{f}$ within the envelope, and Figure 2(c) shows the strategy from Lemma 2.

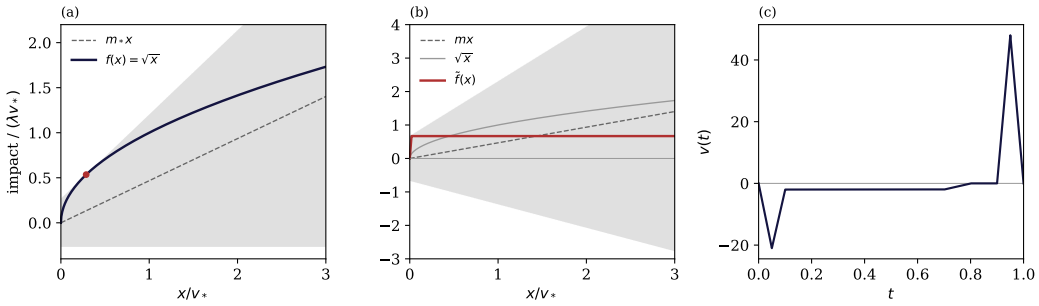


Figure 2. $G(t) = t^{-1/4}$, $\lambda = v_* = 1$, $\eta = \varsigma = s$ with $8s^2 = \nu_G c_G$. (a) Square-root law, chord m_*x, and the band $m_*x \pm (\eta x/\nu_G + \varsigma/c_G)$; the dot marks the contact point. (b) The impact function $\tilde{f}$ of Theorem 2 with $m = m_*$, $\theta = 2.5$, $\delta = 0.02$, and the band $m_*x \pm \theta(\eta x/\nu_G + \varsigma/c_G)$. (c) The round trip of Lemma 2 with $h = 0.1$, $\tau = 0.8$, $\zeta = 2$.

Corollary 1. *Let G be a propagator of positive type with $G \in L^1([0,1])$ and $c_G > 0$, let $\eta, \varsigma > 0$, and let $f = f(\,\cdot\,; 1/2)$ be the square-root law. If*

$$32\,\eta\varsigma < \lambda^2 v_*\,\nu_G c_G, \tag{43}$$

then for every $m \geq 0$ there is a nondecreasing concave instantaneous impact function $\tilde{f}$ with $\Theta^*_m(\tilde{f}) \leq \Theta^*_m(f)$ for which the model admits price manipulation. For square-root impact, any criterion based solely on an envelope threshold can lower the friction product $\eta\varsigma$ required by Theorem 1 by at most a factor of four, even with the best choice of slope m.

Proof. Fix $m \geq 0$ and put $\theta := \Theta^*_m(f)$. By (34) and (43),

$$\theta \;\geq\; \inf_{m' \geq 0} \Theta^*_{m'}(f) = \frac{\lambda\sqrt{\nu_G c_G v_*}}{2\sqrt{2\eta\varsigma}} > 2 \;\geq\; \frac{c_G}{\mu_G},$$

the last step by Lemma 2, so Theorem 2 applies with this θ. If θ_0 is an envelope threshold at slope m that certifies f, then $\theta_0 \geq \Theta^*_m(f) > 2$, contradicting Theorem 2. Thus every such criterion requires $\eta\varsigma \geq \lambda^2 v_* \nu_G c_G/32$. Comparing this with the sufficient bound $\eta\varsigma \geq \lambda^2 v_* \nu_G c_G/8$ in (35) gives the factor of four. □

3.4. The exact threshold for power-law impact

For power-law impact, scaling a fixed round trip reduces the search for a profitable trade to a one-variable minimization. The remaining optimization over trading profiles defines Ξ_ε below. We derive the exact slippage condition in terms of this constant, bound it using explicit round trips, and evaluate it for permanent square-root impact.

Definition 5. *Let $0 < \varepsilon < 1$ and let $f = f(\,\cdot\,; \varepsilon)$ be as in (29). The* extremal manipulation constant *is*

$$\Xi_\varepsilon := \sup\left\{ \frac{-\langle v, \mathcal{P}f(v)\rangle}{\|v\|_2^{2(1-\varepsilon)}\,\|v\|_1^{\varepsilon}} \;:\; v \in C([0,1]) \setminus \{0\},\; \int_0^1 v(t)\,dt = 0 \right\}. \tag{44}$$

The numerator in (44) is homogeneous of degree $2 - \varepsilon$ in v, and the exponents in the denominator are chosen to match, so the quotient is invariant under $v \mapsto cv$ for every $c > 0$. Thus the optimization defining Ξ_ε is over round-trip shapes, with amplitude factored out. The units of Ξ_ε are those of $\lambda v_*^{\varepsilon}$.

Theorem 3. *Let G be a propagator of positive type with $G \in L^1([0,1])$ and $c_G > 0$, let $0 < \varepsilon < 1$, let $\eta, \varsigma > 0$, and let $f = f(\,\cdot\,;\varepsilon)$. The model admits no price manipulation if and only if*

$$\eta^{1-\varepsilon}\varsigma^{\varepsilon} \;\geq\; \varepsilon^{\varepsilon}(1-\varepsilon)^{1-\varepsilon}\,\Xi_{\varepsilon}. \tag{45}$$

Moreover

$$0 \leq \Xi_{\varepsilon} \;\leq\; \frac{\nu_G^{1-\varepsilon} c_G^{\varepsilon} \lambda v_*^{\varepsilon}}{2^{1-\varepsilon}}. \tag{46}$$

Proof. Fix a nonzero round trip v and set

$$A := \|v\|_2^2, \qquad B := \|v\|_1, \qquad K := -\langle v, \mathcal{P}f(v)\rangle.$$

By (29) and oddness, $f(cx;\varepsilon) = c^{1-\varepsilon} f(x;\varepsilon)$ for every $c > 0$ and every $x \in \mathbb{R}$. Each cv with $c > 0$ is again a round trip, and (6) gives

$$C_f^{\eta,\varsigma}[cv] = \eta A c^2 + \varsigma B c - K c^{2-\varepsilon}. \tag{47}$$

The family $\{cv : c > 0\}$ consists of trades with the same timing and relative rates as v, at different volumes. We first determine when one of these trades has negative cost, then optimize over v.

For $K \leq 0$ the right-hand side of (47) is positive for every $c > 0$. Assume $K > 0$ and put

$$\phi(c) := \frac{C_f^{\eta,\varsigma}[cv]}{c} = \eta A c + \varsigma B - K c^{1-\varepsilon}, \qquad c > 0.$$

The derivative $\phi'(c) = \eta A - (1-\varepsilon) K c^{-\varepsilon}$ increases strictly from $-\infty$ to $\eta A > 0$ as c runs from zero to infinity. Thus ϕ decreases until $\phi'(c) = 0$ and increases thereafter. The global minimum is attained at

$$c_0 = \left(\frac{(1-\varepsilon)K}{\eta A}\right)^{1/\varepsilon}.$$

At c_0 one has $K c_0^{1-\varepsilon} = \eta A c_0/(1-\varepsilon)$, so

$$\begin{aligned} \min_{c>0} \phi(c) &= \varsigma B + \eta A c_0 \left(1 - \frac{1}{1-\varepsilon}\right) \\ &= \varsigma B - \frac{\varepsilon}{1-\varepsilon} \eta A c_0. \end{aligned} \tag{48}$$

The cost (47) is therefore negative for some $c > 0$ exactly when $\varsigma B < \varepsilon \eta A c_0/(1-\varepsilon)$. Substituting c_0 and raising both sides to the power ε puts this last inequality in the form

$$\eta^{1-\varepsilon}\varsigma^{\varepsilon} A^{1-\varepsilon} B^{\varepsilon} < \varepsilon^{\varepsilon}(1-\varepsilon)^{1-\varepsilon} K.$$

Dividing by $A^{1-\varepsilon}B^{\varepsilon}$ and taking the supremum over nonzero round trips gives (45).

To prove $\Xi_{\varepsilon} \geq 0$, let $w(t) := \sin(2\pi t)$ on $[0,1]$. For $0 < h < 1$, define the continuous round trip $w_h(t) := w(t/h)$ on $[0,h]$ and $w_h(t) := 0$ on $(h,1]$. Young's inequality on $[0,h]$, followed by a change of time variable, gives

$$\frac{|\langle w_h, \mathcal{P}f(w_h)\rangle|}{\|w_h\|_2^{2(1-\varepsilon)} \|w_h\|_1^{\varepsilon}} \leq \frac{\|w\|_2 \|f(w)\|_2}{\|w\|_2^{2(1-\varepsilon)} \|w\|_1^{\varepsilon}} \int_0^h G(s)\,ds \longrightarrow 0 \quad \text{as } h \downarrow 0.$$

The supremum in (44) is therefore at least zero.

For the upper bound in (46), write

$$\Lambda := \frac{\nu_G^{1-\varepsilon} c_G^{\varepsilon} \lambda v_*^{\varepsilon}}{2^{1-\varepsilon}}$$

and suppose $\Xi_\varepsilon > \Lambda$. Choose $\eta, \varsigma > 0$ with

$$\varepsilon^{\varepsilon}(1-\varepsilon)^{1-\varepsilon}\Lambda \le \eta^{1-\varepsilon}\varsigma^{\varepsilon} < \varepsilon^{\varepsilon}(1-\varepsilon)^{1-\varepsilon}\Xi_\varepsilon.$$

The left inequality is (32), so Theorem 1 excludes price manipulation, while the right inequality together with (45) produces a round trip of negative cost. Hence $\Xi_\varepsilon \le \Lambda$. □

Corollary 2. *Let G, ε, and f be as in Theorem 3, and suppose $\Xi_\varepsilon > 0$. Under rate-only slippage ($\eta > 0$, $\varsigma = 0$) and under spread-only slippage ($\eta = 0$, $\varsigma > 0$) the model admits price manipulation.*

Proof. Since $\Xi_\varepsilon > 0$, choose a round trip v with $K := -\langle v, \mathcal{P}f(v)\rangle > 0$, and set $A := \|v\|_2^2$ and $B := \|v\|_1$. Under rate-only slippage, (47) gives $C_f^{\eta,0}[cv] = \eta A c^2 - K c^{2-\varepsilon} < 0$ for sufficiently small $c > 0$. Under spread-only slippage, $C_f^{0,\varsigma}[cv] = \varsigma B c - K c^{2-\varepsilon} < 0$ for sufficiently large c. □

The conclusion that either form of slippage alone permits manipulation depends on $\Xi_\varepsilon > 0$. For square-root impact, we establish this positivity analytically for $G \equiv 1$ and by a certified numerical construction for $G(t) = t^{-1/4}$. For a general admissible kernel, the corollary remains conditional on $\Xi_\varepsilon > 0$. If $\Xi_\varepsilon = 0$, definition (44) implies that the impact cost is nonnegative for every continuous round trip. Adding rate slippage, spread slippage, or both with nonnegative coefficients therefore preserves nonnegative round-trip cost.

The bound (46) controls Ξ_ε from above. A lower bound on Ξ_ε follows by evaluating the quotient in (44) at an explicit round trip. Consider a path with two constant-rate blocks: trade at a constant rate in one direction over $[0,\tau]$, then at a constant rate in the other direction over $(\tau,1]$. Write

$$\Gamma(x) := \int_0^x G(u)\,du, \qquad I(\ell) := \int_0^\ell \Gamma(x)\,dx = \int_0^\ell (\ell - u)\,G(u)\,du, \tag{49}$$

so that $I(\ell)$ is the self-impact integral of a unit-rate block of length ℓ, and $I(1) - I(\tau) - I(1-\tau)$ is the cross integral $\int_\tau^1 \int_0^\tau G(t-s)\,ds\,dt$.

Proposition 2. *Let G be a propagator of positive type with $G \in L^1([0,1])$ and $c_G > 0$, let $0 < \varepsilon < 1$, and let $f = f(\,\cdot\,;\varepsilon)$. For every $\tau \in (0,1)$,*

$$\Xi_\varepsilon \;\ge\; \frac{\lambda v_*^{\varepsilon}}{2^{\varepsilon}}\big(\tau(1-\tau)\big)^{1-\varepsilon}\left[\frac{I(1) - I(\tau) - I(1-\tau)}{\tau^{1-\varepsilon}(1-\tau)} - \frac{I(\tau)}{\tau^{2-\varepsilon}} - \frac{I(1-\tau)}{(1-\tau)^{2-\varepsilon}}\right]. \tag{50}$$

For $G(t) = t^{-\gamma}$ with $0 < \gamma < 1$, $I(\ell) = \ell^{2-\gamma}/((1-\gamma)(2-\gamma))$, and (50) reads

$$\Xi_\varepsilon \;\ge\; \frac{\lambda v_*^{\varepsilon}\,\big(\tau(1-\tau)\big)^{1-\varepsilon}}{2^{\varepsilon}(1-\gamma)(2-\gamma)}\left[\frac{1 - \tau^{2-\gamma} - (1-\tau)^{2-\gamma}}{\tau^{1-\varepsilon}(1-\tau)} - \tau^{\varepsilon-\gamma} - (1-\tau)^{\varepsilon-\gamma}\right]. \tag{51}$$

Proof. Let $v := -\tau^{-1}$ on $[0,\tau]$ and $v := (1-\tau)^{-1}$ on $(\tau,1]$, so that $\int_0^1 v = 0$, $\|v\|_1 = 2$, and $\|v\|_2^2 = 1/(\tau(1-\tau))$. By oddness of f and $f(x;\varepsilon) = \lambda v_*^{\varepsilon} x^{1-\varepsilon}$ for $x > 0$, splitting $\langle v, \mathcal{P}f(v)\rangle$ over the four pairs of blocks gives

$$-\langle v, \mathcal{P}f(v)\rangle = \lambda v_*^{\varepsilon}\left[\frac{I(1) - I(\tau) - I(1-\tau)}{(1-\tau)\,\tau^{1-\varepsilon}} - \frac{I(\tau)}{\tau^{2-\varepsilon}} - \frac{I(1-\tau)}{(1-\tau)^{2-\varepsilon}}\right],$$

where the three terms are the cross pair, the self pair on $[0,\tau]$, and the self pair on $(\tau,1]$; the pair with the sells after the buys is empty. Dividing by

$$\|v\|_2^{2(1-\varepsilon)}\|v\|_1^{\varepsilon} = \frac{2^{\varepsilon}}{\big(\tau(1-\tau)\big)^{1-\varepsilon}}$$

gives the right-hand side of (50). The path v is a step function. Replace the jump by a linear ramp on $[\tau-\delta/2,\tau+\delta/2]$, with $0<\delta<2\min\{\tau,1-\tau\}$. The centered ramp preserves $\int_0^1 v = 0$. As $\delta\downarrow 0$, the ramped paths converge to v in L^1 with a uniform bound, so their quotients in (44) converge to that of v, and (50) follows. Substituting $I(\ell)=\ell^{2-\gamma}/((1-\gamma)(2-\gamma))$ gives (51). □

Combined with Theorem 3, (50) is an explicit necessary condition: the model admits price manipulation whenever $\eta^{1-\varepsilon}\varsigma^{\varepsilon}$ falls below $\varepsilon^{\varepsilon}(1-\varepsilon)^{1-\varepsilon}$ times the right-hand side of (50) for some τ. Taking the supremum over τ gives the strongest lower bound on Ξ_ε within this two-block family. Allowing the rate to vary within each trading phase can improve that bound, as Figure 3 illustrates for square-root impact with $G(t)=t^{-1/4}$. The exact threshold for that transient kernel remains open. For permanent impact, Proposition 3 determines $\Xi_{1/2}$ by optimizing over all round trips.

Remark 2. *For square-root impact,* (45) *and the nonnegativity in* (46) *give the exact condition* $4\eta\varsigma \geq \Xi_{1/2}^2$. *The sufficient condition* (35) *replaces* $\Xi_{1/2}$ *by its upper bound from* (46).

For $G(t)=t^{-1/4}$ and $\lambda = v_* = 1$, Figure 3 illustrates how slippage suppresses the profit of a numerically computed round trip v_{num}. The numerical search produces a trading rate that is constant on each of 128 equal time intervals. We subtract its average rate from every interval's rate so that total purchases equal total sales exactly. Its impact cost is evaluated using explicit, exact integrals of the kernel over pairs of time intervals, including the singular diagonal cells. Rational interval arithmetic bounds all numerical evaluation errors and certifies the quotient q_{num} in (44):

$$0.042012615472 < q_{\mathrm{num}} < 0.042012615473.$$

Replacing the jumps by symmetric linear ramps preserves zero total volume. Explicit bounds on the resulting changes in impact cost and the two norms certify a continuous round trip with quotient greater than 0.0420125. Consequently,

$$\Xi_{1/2} > 0.0420125.$$

The accompanying script `certify_round_trip_lower_bound.py` verifies these inequalities using exact rational arithmetic and integer root bounds. The proof requires only this verified candidate, irrespective of whether the numerical search finds a global optimum. The exact value of $\Xi_{1/2}$ for this kernel remains open. With $\eta=\varsigma=s$, some multiple cv_{num}, $c>0$, has negative cost exactly when $s<s_{\mathrm{ray}} := q_{\mathrm{num}}/2$. Panel (b) shows the cost as the scale c varies, for slippage below, at, and above this threshold.

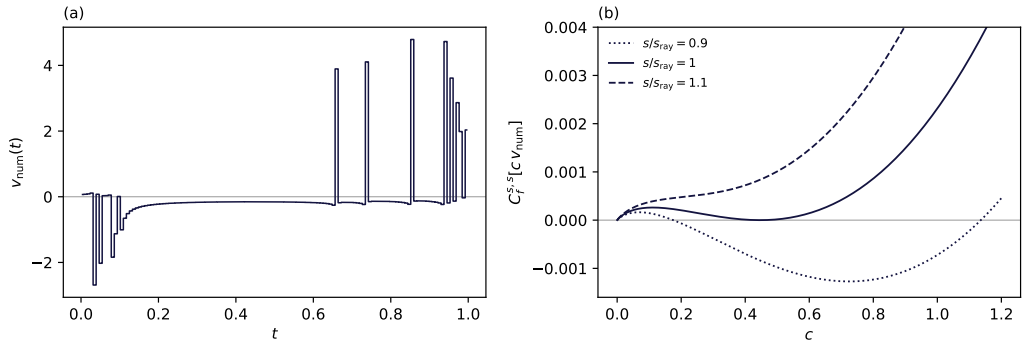


Figure 3. (a) Round trip v_{num} returned by multistart local optimization of (44) with $G(t) = t^{-1/4}$, $\varepsilon = 1/2$, $\lambda = v_* = 1$, 128 equal time buckets, normalized to $\|v_{\text{num}}\|_2 = 1$. (b) Cost (47) along the ray through v_{num} at $\eta = \varsigma = s$, for $s/s_{\text{ray}} \in \{0.9, 1, 1.1\}$ with $s_{\text{ray}} = q_{\text{num}}/2$.

3.4.1. Permanent impact

For $G \equiv 1$ the transient model reduces to permanent impact, the setting of Huberman and Stanzl (2004), with $c_G = 1$ and $\nu_G = 1/2$, so that (46) gives $\Xi_{1/2} \le \lambda\sqrt{v_*}/2$. For permanent impact, the numerator $-\langle v, \mathcal{P}f(v)\rangle$ in (44) can be expressed using the trader's position. For a round trip v, write $Z(t) := \int_0^t v(s)\,ds$ for the position, so that $Z(0) = Z(1) = 0$, and $F(t) := \int_0^t f(v(s))\,ds$ for the accumulated impact. Integrating by parts,

$$\begin{aligned} -\langle v, \mathcal{P}f(v)\rangle &= -\int_0^1 Z'(t)F(t)\,dt \\ &= \int_0^1 Z(t)\,f\big(v(t)\big)\,dt. \end{aligned} \tag{52}$$

Thus the profit of a round trip under permanent impact is the position held, weighted by the impact of the current trade. The sharp constant for square-root impact has the following algebraic expression.

Proposition 3. *Let $G \equiv 1$ and $\varepsilon = 1/2$, and let z_* be the unique root in $(0,1)$ of*

$$18z^3 + 39z^2 + 8z - 5 = 0. \tag{53}$$

Then

$$\Xi_{1/2} = \lambda\sqrt{v_*}\sqrt{\frac{1584z_*^2 + 2046z_* - 535}{15360}} \approx 0.081\,\lambda\sqrt{v_*}. \tag{54}$$

The supremum $\Xi_{1/2}$ is approached by continuous round trips with a single buy phase followed by a sell phase.

The proof of Proposition 3 is given in Appendix A.

For permanent square-root impact, Theorem 3 therefore gives the exact no-manipulation condition

$$4\eta\varsigma \ge \lambda^2 v_*\,\frac{1584z_*^2 + 2046z_* - 535}{15360}.$$

3.5. Transaction-triggered price manipulation

Transaction-triggered price manipulation occurs when trading against the direction of a prescribed order lowers its execution cost (Alfonsi et al., 2012). For a buy order, for example, allowing intermediate sales can lower the cost below the infimum over buy-only strategies. In a discrete-time model with linear permanent and temporary impact, Benazzoli

and Di Persio (2017) study optimal liquidation both with unrestricted trade signs and with a pure-selling constraint, allowing stochastic drift and temporary-impact coefficients. Here we seek conditions under which restricting trades to the prescribed direction preserves the infimum of execution cost. For $Q \in \mathbb{R} \setminus \{0\}$, define

$$\mathcal{A}_Q := \left\{ v \in C([0,1]) \;\middle|\; \int_0^1 v(t)\,dt = Q \right\}$$

and its one-sided subclass

$$\mathcal{A}_Q^{\text{one}} := \{ v \in \mathcal{A}_Q \mid Qv(t) \geq 0 \text{ for every } t \in [0,1] \}.$$

Definition 6. *A model is said to* admit transaction-triggered price manipulation *if, for some* $Q \in \mathbb{R} \setminus \{0\}$,

$$\inf_{v \in \mathcal{A}_Q} C_f^{\eta,\varsigma}[v] < \inf_{v \in \mathcal{A}_Q^{\text{one}}} C_f^{\eta,\varsigma}[v]. \tag{55}$$

For a globally Lipschitz impact function f*, set*

$$L_f := \sup_{x \neq y} \frac{|f(x) - f(y)|}{|x - y|}. \tag{56}$$

The constant L_f has the same units as η and λ. For a propagator $G \in L^1([0,1])$, define

$$R_G := \max_{0 \leq t \leq 1} \left[\int_0^t G(r)\,dr + \int_0^{1-t} G(r)\,dr \right]. \tag{57}$$

The two integrals measure the kernel mass before and after time t. Their sum is continuous in t, so the maximum exists, and $R_G \leq 2c_G$. If G is nonincreasing, the integral $x \mapsto \int_0^x G(r)\,dr$ is concave. Its values at t and $1-t$ therefore have their largest sum at $t = 1/2$, giving

$$R_G = 2 \int_0^{1/2} G(r)\,dr.$$

Theorem 4. *Let* G *be a propagator with* $G \in L^1([0,1])$*, and let* f *be a globally Lipschitz instantaneous impact function. Suppose*

$$\eta \geq L_f R_G. \tag{58}$$

For every $Q \in \mathbb{R} \setminus \{0\}$ *and every* $v \in \mathcal{A}_Q$*, there is a* $u \in \mathcal{A}_Q^{\text{one}}$ *such that, with*

$$H := \|v - u\|_1, \qquad D := \|v\|_2^2 - \|u\|_2^2,$$

one has $D \geq 0$ *and*

$$C_f^{\eta,\varsigma}[v] - C_f^{\eta,\varsigma}[u] \geq (\eta - L_f R_G) D + \varsigma H. \tag{59}$$

Consequently,

$$\inf_{v \in \mathcal{A}_Q} C_f^{\eta,\varsigma}[v] = \inf_{v \in \mathcal{A}_Q^{\text{one}}} C_f^{\eta,\varsigma}[v] \tag{60}$$

for every $Q \in \mathbb{R} \setminus \{0\}$*. If* v *changes sign and either* $\eta > L_f R_G$ *or* $\varsigma > 0$*, then* $D > 0$ *and* $H > 0$*, so the right-hand side of* (59) *is positive and* $C_f^{\eta,\varsigma}[v] > C_f^{\eta,\varsigma}[u]$*.*

Proof. Write $J_f[w] := \langle w, \mathcal{P} f(w) \rangle$ for the impact cost. We first bound its change between two trading-rate paths x and y satisfying $\|x\|_\infty, \|y\|_\infty \leq M$. The identity

$$J_f[x] - J_f[y] = \langle x - y, \mathcal{P} f(x) \rangle + \langle \mathcal{P}^* y, f(x) - f(y) \rangle \tag{61}$$

and the bounds $|f(x(t))| \le L_f M$ and $|f(x(t)) - f(y(t))| \le L_f |x(t) - y(t)|$ give

$$|J_f[x] - J_f[y]| \le L_f M \int_0^1 |x(t) - y(t)| \left[\int_0^t G(r)\, dr + \int_0^{1-t} G(r)\, dr \right] dt$$
$$\le L_f R_G M \|x - y\|_1. \tag{62}$$

Oddness of f gives $C_f^{\eta,\varsigma}[-v] = C_f^{\eta,\varsigma}[v]$, so it suffices to consider $Q > 0$. Define

$$p(t) := \max\{v(t), 0\}, \qquad n(t) := \max\{-v(t), 0\}, \qquad B := \int_0^1 n(t)\, dt.$$

If $B = 0$, take $u = v$. Suppose $B > 0$. We remove equal volumes from the positive and negative parts by progressively reducing their largest rates. At removal volume $s \in [0, B]$, choose caps $a(s) \in [0, \|p\|_\infty]$ and $b(s) \in [0, \|n\|_\infty]$ such that

$$\int_0^1 \min\{p(t), a(s)\}\, dt = Q + B - s,$$
$$\int_0^1 \min\{n(t), b(s)\}\, dt = B - s. \tag{63}$$

For a continuous nonnegative function h with positive maximum, the map $c \mapsto \int_0^1 \min\{h(t), c\}\, dt$ is continuous and strictly increasing on $[0, \|h\|_\infty]$. Indeed, if $0 \le c < d \le \|h\|_\infty$, continuity gives a set of positive measure on which $h > c$, and increasing the cap from c to d strictly increases the integral. Thus the caps in (63) exist uniquely and are continuous and decreasing, with $a(0) = \|p\|_\infty$ and $b(0) = \|n\|_\infty$.

Define the intermediate trading-rate path

$$w_s(t) := \min\{p(t), a(s)\} - \min\{n(t), b(s)\}.$$

Every w_s is continuous and has total volume Q. Moreover, $w_0 = v$ and $b(B) = 0$, so

$$u := w_B = \min\{p, a(B)\} \in \mathcal{A}_Q^{\text{one}}.$$

For $0 \le s < r \le B$, the positive and negative parts decrease on disjoint sets, and each loses volume $r - s$. Consequently,

$$\|w_s - w_r\|_1 = 2(r - s), \qquad \|w_s\|_\infty = \max\{a(s), b(s)\}. \tag{64}$$

In particular,

$$H = \|v - u\|_1 = 2B = \|v\|_1 - \|u\|_1. \tag{65}$$

To calculate the saving in quadratic slippage, consider the positive part. Wherever its value changes between removal volumes s and r, the sum of its old and new values lies between $2a(r)$ and $2a(s)$. Factoring the difference of their squares and using (63) therefore gives

$$2a(r)(r - s) \le \|\min\{p, a(s)\}\|_2^2 - \|\min\{p, a(r)\}\|_2^2$$
$$\le 2a(s)(r - s).$$

The same inequalities hold for the negative part with b in place of a. Sum over consecutive points of a partition of $[0, B]$ and let its mesh tend to zero. Continuity of the caps yields

$$D = \|v\|_2^2 - \|u\|_2^2 = 2 \int_0^B [a(s) + b(s)]\, ds. \tag{66}$$

Since $a(s), b(s) > 0$ for $s < B$, one has $D > 0$.

We now compare the impact costs along the same partition. The caps are decreasing, so $\max\{a(s), b(s)\}$ bounds the absolute rates of both w_s and w_r for $s < r$. Applying (62) and (64) to each successive pair, summing, and taking the limit gives

$$\begin{aligned} |J_f[v] - J_f[u]| &\le 2L_f R_G \int_0^B \max\{a(s), b(s)\}\, ds \\ &\le 2L_f R_G \int_0^B [a(s) + b(s)]\, ds \\ &= L_f R_G D. \end{aligned} \tag{67}$$

The local slippage saving is exactly $\eta D + \varsigma H$ by (65) and (66). Combining this saving with (67) proves (59).

We have constructed, for each $v \in \mathcal{A}_Q$, a path $u \in \mathcal{A}_Q^{\text{one}}$ satisfying $C_f^{\eta,\varsigma}[u] \le C_f^{\eta,\varsigma}[v]$. Hence

$$\inf_{v \in \mathcal{A}_Q} C_f^{\eta,\varsigma}[v] \ge \inf_{v \in \mathcal{A}_Q^{\text{one}}} C_f^{\eta,\varsigma}[v].$$

The inclusion $\mathcal{A}_Q^{\text{one}} \subset \mathcal{A}_Q$ gives the reverse inequality, proving (60). □

For linear impact $f(v) = \lambda v$, Alfonsi et al. (2012) prove that convex, nonincreasing kernels yield one-sided optimal strategies in their discrete-time model, and discuss the extension to continuous time. This excludes transaction-triggered price manipulation even when $\eta = 0$. Our sufficient condition (58) specializes to $\eta \ge \lambda R_G$, and is therefore conservative in this setting. The absolute-value estimate (62) controls changes in impact cost using the Lipschitz constant and kernel mass. The resulting bound applies to nonlinear Lipschitz impact and general nonnegative integrable kernels; the sharper linear-impact result exploits convexity and monotonicity of the kernel.

3.6. A globally Lipschitz approximation

For $0 < \varepsilon < 1$, the power law (29) has

$$\frac{f(x;\varepsilon)}{x} = \lambda \left(\frac{x}{v_*}\right)^{-\varepsilon} \longrightarrow +\infty \qquad \text{as } x \downarrow 0,$$

so its global Lipschitz constant equals $+\infty$. A linear core gives a globally Lipschitz approximation while preserving the calibration at v_*. For a core rate $v_c > 0$, define

$$f_c(x;\varepsilon) := \lambda x \left(\frac{x^2 + v_c^2}{v_*^2 + v_c^2}\right)^{-\varepsilon/2}, \qquad x \in \mathbb{R}. \tag{68}$$

The function $f_c(\,\cdot\,;\varepsilon)$ is odd, satisfies $f_c(v_*;\varepsilon) = \lambda v_*$, and obeys

$$f_c(x;\varepsilon) = \lambda \left(\frac{v_*^2 + v_c^2}{v_c^2}\right)^{\varepsilon/2} x + o(x) \qquad \text{as } x \to 0.$$

Its derivative is

$$f_c'(x;\varepsilon) = \lambda \left(v_*^2 + v_c^2\right)^{\varepsilon/2} \frac{v_c^2 + (1-\varepsilon)x^2}{\left(x^2 + v_c^2\right)^{1+\varepsilon/2}}.$$

This derivative is positive and decreases with $|x|$, so $L_{f_c} = f_c'(0;\varepsilon)$, that is,

$$L_{f_c} = \lambda \left(\frac{v_*^2 + v_c^2}{v_c^2}\right)^{\varepsilon/2}. \tag{69}$$

For each $x \neq 0$, $f_c(x;\varepsilon)$ converges to the odd extension of (29) as $v_c \downarrow 0$.

Corollary 3. *Let G be a propagator with $G \in L^1([0,1])$, let $0 < \varepsilon < 1$, let $v_c > 0$, and let $f = f_c(\cdot;\varepsilon)$. If*

$$\eta \geq \lambda R_G \left(\frac{v_*^2 + v_c^2}{v_c^2} \right)^{\varepsilon/2}, \tag{70}$$

then (60) holds for every $Q \in \mathbb{R} \setminus \{0\}$.

For $c_G > 0$, set $\rho := v_c / v_*$. In terms of the dimensionless ratio $\eta/(c_G\lambda)$, condition (70) becomes

$$\frac{\eta}{c_G \lambda} \geq \frac{R_G}{c_G} \left(1 + \rho^{-2}\right)^{\varepsilon/2}. \tag{71}$$

Both ρ and R_G/c_G are dimensionless, and R_G/c_G is invariant under the gauge freedom of Section 2.1. For the power-law kernel $G(t) = t^{-\gamma}$ with $0 < \gamma < 1$,

$$c_G = \frac{1}{1-\gamma}, \qquad R_G = \frac{2^\gamma}{1-\gamma},$$

so (71) reduces to

$$\frac{\eta}{c_G \lambda} \geq 2^\gamma \left(1 + \rho^{-2}\right)^{\varepsilon/2}. \tag{72}$$

The lower bound on $\eta/(c_G\lambda)$ in (72) grows as $\rho = v_c/v_*$ decreases to zero. For fixed ρ, the kernel factor 2^γ increases from 1 to 2 as γ ranges from 0 to 1.

4. Conclusion

Gatheral (2010) showed that nonlinear instantaneous impact can permit profitable round trips. Square-root impact with the kernel $G(t) = t^{-1/4}$ provides one example. The execution model studied here also charges a fixed half-spread and a rate-proportional concession. Theorem 1 gives a sufficient condition for these slippage costs to offset the profit from impact. For a comparison slope $m \geq 0$, the condition is $\Theta_m^* \leq 1$, where Θ_m^* measures the deviation $|f(x) - mx|$ relative to $\eta x/\nu_G + \varsigma/c_G$. If $\Theta_m^* < 1$, every nonzero round trip has strictly positive expected cost.

The half-spread controls manipulation at small trading rates, while the rate-proportional concession controls manipulation at large rates. For power-law impact with $0 < \varepsilon < 1$ and $\Xi_\varepsilon > 0$, Corollary 2 shows that either form of slippage alone permits manipulation. Together, the two forms of slippage rule out manipulation under (32). For square-root impact, the sufficient condition is $8\eta\varsigma \geq \lambda^2 v_* \nu_G c_G$. The equivalent bound (38) uses the empirical impact coefficient Y_0, daily volatility σ_d, the quoted spread s_{BA}, and the kernel ratio ν_G/c_G to bound the execution-cost coefficient b_η.

For power-law impact, homogeneity reduces the exact no-manipulation condition to the constant Ξ_ε in Theorem 3. The exact condition (45) provides a benchmark for the sufficient bound in Theorem 1. For square-root impact, criteria based on an envelope threshold can improve that bound by at most a factor of four in $\eta\varsigma$. The envelope also permits impact functions that produce substantial price changes at arbitrarily small trading rates. Sharper criteria can use the specific dependence of square-root impact on the trading rate.

For permanent square-root impact, Proposition 3 determines $\Xi_{1/2}$ exactly, as an algebraic number, and with it the exact boundary in $\eta\varsigma$. The extremal round trip accumulates a position slowly at a falling rate and unwinds it quickly. The proof first rearranges piecewise constant round trips into a single buy phase followed by a sell phase, preserving the slippage costs and weakly increasing the profit from impact. A pointwise inequality then

identifies the optimal rate profile, and approximation extends the bound to continuous round trips. For $G(t) = t^{-1/4}$, the numerical round trip of Figure 3 provides a lower bound on $\Xi_{1/2}$; determining $\Xi_{1/2}$ for this kernel remains open.

The spectral constant $\nu_G = \|\mathcal{G}\|/2$ weights the rate-proportional slippage in Θ_m^*, while the kernel mass c_G weights the half-spread. Proposition 1 gives a further cost bound with penalty $\lambda(a_G\kappa_v + \nu_G\kappa_v^2/4)$. For small relative deviation κ_v, the penalty is governed by $\lambda a_G\kappa_v$. Relating the ratio a_G/ν_G to empirical features of market impact is a question for further study.

Theorem 4 rules out transaction-triggered manipulation for integrable propagators under $\eta \geq L_f R_G$. Progressively capping the buying and selling rates removes equal volumes from both sides until the strategy trades only in the prescribed direction, reducing expected cost. The approximation f_c in (68) gives square-root impact a finite Lipschitz constant by making impact approximately linear near zero. Using more of the shape of f may improve the bound $L_f R_G$ on the required slippage coefficient η.

Data Availability Statement: The scripts used to generate Figure 1, Figure 2, Figure 3, Figure A1, and Table 1, the evaluation of the sharp constant in Proposition 3, and the certificates named in the text are available at https://github.com/gordonritter/no-manipulation-with-slippage.

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Appendix A. Sharp constant for permanent impact

Proof of Proposition 3. Take $\lambda = v_* = 1$. For a rate v, write $Z(t) := \int_0^t v(s)\,ds$, and put $\phi(x) := \operatorname{sign}(x)\sqrt{|x|}$, $A(v) := \|v\|_2^2$, $B(v) := \|v\|_1$, and $K(v) := \int_0^1 Z(t)\phi(v(t))\,dt$. For a round trip, $Z(0) = Z(1) = 0$, so integration by parts gives $K(v) = -\langle v, \mathcal{P}\phi(v)\rangle$. Hence $\Xi_{1/2}$ is the supremum of $K(v)/\sqrt{A(v)B(v)}$ over nonzero continuous rates v with $\int_0^1 v(t)\,dt = 0$.

We first explain why an upper bound for piecewise constant round trips also bounds this supremum. Fix any such continuous rate v. Divide $[0,1]$ into n equal intervals and replace v on each interval by its average there, defining

$$v_n(t) := n\int_{(j-1)/n}^{j/n} v(s)\,ds, \qquad \frac{j-1}{n} \leq t < \frac{j}{n}, \qquad j = 1, \ldots, n.$$

Set $v_n(1)$ equal to the value on the last interval. The total signed volume is preserved: $\int_0^1 v_n(t)\,dt = \int_0^1 v(t)\,dt = 0$. Thus each v_n is a piecewise constant round trip. Uniform continuity of v implies $v_n \to v$ uniformly and therefore in L^2.

To pass the manipulation quotient to the limit, observe that for any sequence $u_n \to u$ in $L^2([0,1])$, the estimate $|\phi(a) - \phi(b)|^2 \leq 2|a-b|$ gives

$$\|\phi(u_n) - \phi(u)\|_2^2 \leq 2\|u_n - u\|_1 \leq 2\|u_n - u\|_2 \longrightarrow 0.$$

Boundedness of $\mathcal{P}$ then gives $K(u_n) \to K(u)$, while convergence in L^2 also gives $A(u_n) \to A(u)$ and $B(u_n) \to B(u)$. If u is nonzero, $A(u)B(u) > 0$, so the corresponding quotients converge as well. Applying this to the interval averages v_n, any common upper bound for the quotients of nonzero piecewise constant round trips also bounds the quotient of v. Since v was arbitrary, the same bound holds for $\Xi_{1/2}$.

We next construct a family of candidate rates v_r. We show that each candidate maximizes K among round trips with the same A and B, then maximize the quotient over the family. For $r \in (0,1)$, let $\rho = \rho(r) > 1$ be the solution of

$$\frac{1-r^4}{r} = \frac{\rho^4-1}{\rho}, \tag{A1}$$

and define the following quantities to express the duration, profit, squared-rate integral, and total volume of the construction:

$$\begin{aligned}
\hat{\tau} &:= 6(\rho - r) + \frac{2}{3}(r^{-3} - \rho^{-3}),\\
\hat{K} &:= 4\Big[2(1-r) + \frac{r^{-3}-1}{3} - \frac{3}{5}(1-r^5)\Big] - 4\Big[\frac{3}{5}(\rho^5-1) - 2(\rho-1) - \frac{1-\rho^{-3}}{3}\Big],\\
\hat{A} &:= 2\Big[\frac{3}{5}(1-r^5) + (1-r)\Big] + 2\Big[\frac{3}{5}(\rho^5-1) + (\rho-1)\Big], \qquad \hat{B} := \frac{4(1-r^4)}{r}.
\end{aligned} \tag{A2}$$

Set $\omega := 1/\hat{\tau}$. For each $r \in (0,1)$, construct a position path Z from two parametrized curves in the (t, Z) plane, each traversed as the parameter s decreases. The buy curve is

$$t_{\mathrm{b}}(s) := \omega\Big[6(1-s) + \frac{2}{3}(s^{-3}-1)\Big], \qquad Z_{\mathrm{b}}(s) := \frac{2\omega^3(1-s^4)}{s}, \qquad r \le s \le 1. \tag{A3}$$

Put $T_{\mathrm{b}} := t_{\mathrm{b}}(r)$. The sell curve is

$$t_{\mathrm{s}}(s) := T_{\mathrm{b}} + \omega\Big[6(\rho - s) + \frac{2}{3}(s^{-3} - \rho^{-3})\Big], \qquad Z_{\mathrm{s}}(s) := \frac{2\omega^3(s^4-1)}{s}, \qquad 1 \le s \le \rho. \tag{A4}$$

The time maps decrease, with $t_{\mathrm{b}}(1) = 0$ and $t_{\mathrm{s}}(1) = 1$. The corner equation (A1) gives $Z_{\mathrm{b}}(r) = Z_{\mathrm{s}}(\rho)$, so the curves define a continuous position Z with $Z(0) = Z(1) = 0$. Differentiation gives the rate $v := Z'$:

$$v(t_{\mathrm{b}}(s)) = \omega^2 s^2, \qquad v(t_{\mathrm{s}}(s)) = -\omega^2 s^2.$$

For $j \in \{\mathrm{b}, \mathrm{s}\}$, the substitution $dt_j = -2\omega(3 + s^{-4})\,ds$ yields

$$K(v) = \omega^5 \hat{K}, \qquad A(v) = \omega^5 \hat{A}, \qquad B(v) = \omega^3 \hat{B}.$$

Denote the rate constructed from (A3) and (A4) by v_r. The quotient $K(v_r)/\sqrt{A(v_r)B(v_r)}$ equals $\hat{K}/(\hat{\tau}\sqrt{\hat{A}\hat{B}})$, where the hatted quantities depend on r through (A1) and (A2). Interpolate v_r linearly across $[T_{\mathrm{b}} - \delta, T_{\mathrm{b}} + \delta]$, then subtract the mean of the interpolated rate. The resulting continuous round trips converge to v_r in L^2 as $\delta \downarrow 0$ and have a single sign change for sufficiently small δ. Continuity of $v \mapsto K(v)/\sqrt{A(v)B(v)}$ gives

$$\Xi_{1/2} \ge \sup_{0<r<1} \frac{\hat{K}}{\hat{\tau}\sqrt{\hat{A}\hat{B}}}. \tag{A5}$$

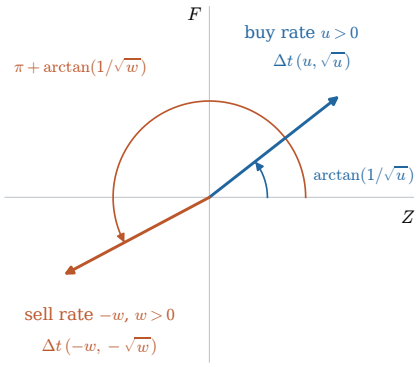
To prove the reverse inequality in (A5), consider an arbitrary piecewise constant round trip v. The path (Z, F), with $F(t) := \int_0^t \phi(v(s))\,ds$, consists of directed edges $\Delta t\,(v, \phi(v))$, where Δt is the duration of a constant-rate interval. Close the path (Z, F) by a vertical edge from $(0, F(1))$ to the origin. The closed path has signed area $\int Z\,dF = K(v)$, with counterclockwise orientation positive, since the closing edge contributes zero.

We use the following convexification result of Fáry and Makai (1982): for a finite collection of directed planar edges whose sum is zero, arranging the edges in counterclockwise angular order gives a closed convex polygon whose signed area is at least that

of any other ordering. The result applies also to self-intersecting polygonal paths; if all edges are collinear, the signed area is zero. Vysotsky (2025) proves the equivalence of the winding-number and line-integral definitions of signed area.

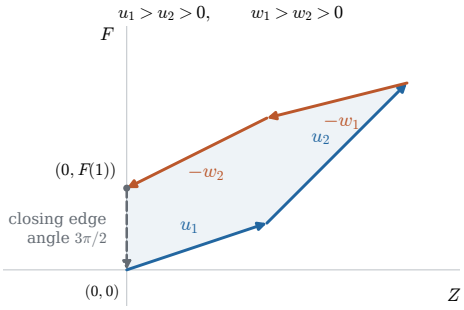
Replacing v by $-v$ changes both Z and F to their negatives and preserves K, A, B, so take $F(1) \geq 0$. Measured counterclockwise from the positive horizontal axis, a buy edge with rate $u > 0$ has angle $\arctan(1/\sqrt{u})$, and a sell edge with rate $-w$, $w > 0$, has angle $\pi + \arctan(1/\sqrt{w})$. Each angle decreases as its rate magnitude increases. Thus increasing angular order places the buys first in decreasing rate, followed by the sells in decreasing magnitude. If $F(1) > 0$, the closing edge has angle $3\pi/2$ and comes last. If $F(1) = 0$, the trading edges already form a closed path and the zero closing edge is omitted. Figure A1 illustrates these edge angles and their ordering.

(a) Buy and sell edge angles



Angles measured counterclockwise from the positive Z axis

(b) Counterclockwise angular order



$F(1) > 0$: the downward closing edge comes last

If $F(1) = 0$, the closing edge has zero length.

Figure A1. Angular ordering in the (Z, F) plane. (a) Buy and sell edges and their counterclockwise angles. (b) Ordered trading edges with $u_1 > u_2 > 0$ and $w_1 > w_2 > 0$. The dashed closing edge has angle $3\pi/2$ when $F(1) > 0$.

Each trading edge retains its rate and duration under this permutation. Repeated rates give edges with the same direction; these may be combined by adding their durations. Zero-rate intervals give zero edges and may be placed at the end of the trading schedule, retaining their total duration. Thus the rearrangement weakly increases K while preserving $A(v), B(v)$ and elapsed time. It suffices to bound $K(v)/\sqrt{A(v)B(v)}$ for paths with one buy phase followed by one sell phase and any remaining idle time.

The quotient $K(v)/\sqrt{A(v)B(v)}$ is invariant under positive amplitude scaling. Scale the rearranged rate v so that $B(v) = 2$; the peak inventory is then 1. Write $A := A(v)$, $B := B(v)$, and $K := K(v)$. For inventory $x \in (0, 1)$, let $a(x)$ be the reciprocal buy rate and $b(x)$ the reciprocal magnitude of the sell rate. These are positive piecewise constant functions on the inventory intervals traversed during trading; their values at the finitely many interval endpoints may be assigned arbitrarily.

To justify the substitution, consider a constant buy rate $u > 0$ on a time interval I over which inventory increases from x_0 to x_1. Since $dx = u\,dt$, its contributions are

$$|I| = \frac{x_1 - x_0}{u}, \qquad \int_I v^2\,dt = u(x_1 - x_0), \qquad \int_I Z\phi(v)\,dt = \int_{x_0}^{x_1} \frac{x}{\sqrt{u}}\,dx.$$

For a constant sell rate $-w$, $w > 0$, inventory decreases from x_1 to x_0, so $dt = -dx/w$ and

$$|I| = \frac{x_1 - x_0}{w}, \qquad \int_I v^2\,dt = w(x_1 - x_0), \qquad \int_I Z\phi(v)\,dt = -\int_{x_0}^{x_1} \frac{x}{\sqrt{w}}\,dx.$$

The buy intervals partition the inventory range $[0,1]$, as do the sell intervals. Setting $a = 1/u$ and $b = 1/w$ on the corresponding intervals and summing gives

$$T_{\mathrm{a}} := \int_0^1 (a+b)\,dx \le 1,$$
$$A = \int_0^1 (a^{-1}+b^{-1})\,dx,$$
$$K = \int_0^1 x(\sqrt{a}-\sqrt{b})\,dx. \tag{A6}$$

Here T_{a} is the time spent trading; idle intervals contribute zero to A, B, K. Each finite piecewise constant path has a positive minimum among its nonzero rate magnitudes, so the substitutions above apply on every trading interval. This minimum may tend to zero along the sequence of interval averages used earlier; the established L^2-continuity of K, A, B then passes the resulting quotient bound to continuous round trips. Cauchy–Schwarz gives $AT_{\mathrm{a}} \ge 4$, hence $A \ge 4$.

To compare paths with fixed A and $B = 2$, subtract penalties for the squared-rate integral and the time spent trading. The penalized objective separates into a buy term and a sell term at each inventory level. Fix $\mu, \ell > 0$. For each $x \in [0,1]$, maximize separately

$$F_+(a;x) := x\sqrt{a} - \frac{\mu}{a} - \ell a, \qquad F_-(b;x) := -x\sqrt{b} - \frac{\mu}{b} - \ell b$$

over positive a, b. Each objective tends to $-\infty$ as its argument tends to zero or infinity. The function $F_+(a;x)$ is strictly concave in a. The derivative $\partial_b F_-(b;x)$, multiplied by b^2, is

$$\mu - \frac{x}{2}b^{3/2} - \ell b^2,$$

which decreases strictly from μ to $-\infty$. Each maximum is therefore unique. Let $a_*(x)$ and $b_*(x)$ be the maximizers, and write their corresponding trading rates as $w_+^2 := 1/a_*$ and $-w_-^2 := -1/b_*$, with $w_+, w_- > 0$. The stationarity equations are

$$\ell = \mu w_+^4 + \frac{x w_+}{2}, \qquad \ell = \mu w_-^4 - \frac{x w_-}{2}. \tag{A7}$$

The rates v_r from (A3) and (A4) satisfy (A7) with $\mu = 1$ and $\ell = \omega^4$ before amplitude normalization. Set $c := 2/(\omega^3 \hat{B})$. Multiplying v_r and its position by c gives $B(cv_r) = 2$ and preserves (A7) with $\mu = c^{-1/2}$ and $\ell = c^{3/2}\omega^4$. Both multipliers are positive.

Fix r, and put $A_* := A(cv_r)$ and $K_* := K(cv_r)$. For any competing buy-then-sell path with $B = 2$ and $A = A_*$, integrate the inequalities $F_+(a;x) \le F_+(a_*;x)$ and $F_-(b;x) \le F_-(b_*;x)$ over $x \in [0,1]$ to obtain

$$K - \mu A - \ell T_{\mathrm{a}} \le K_* - \mu A_* - \ell.$$

Since $T_{\mathrm{a}} \le 1$, it follows that $K \le K_*$. We next verify that the normalized rates cv_r, $0 < r < 1$, attain every squared-rate integral $A > 4$.

Put $s := r\rho$, $d := \rho - r$, and $z := s^2$. Factoring (A1) gives $r\rho(r^2 - r\rho + \rho^2) = 1$, so

$$d^2 = s^{-1} - s, \qquad r = \frac{\sqrt{s^{-1}+3s} - \sqrt{s^{-1}-s}}{2}, \qquad \rho = \frac{\sqrt{s^{-1}+3s} + \sqrt{s^{-1}-s}}{2}.$$

Thus $0 < z < 1$ parameterizes the rates v_r through $z = (r\rho(r))^2$. Substitution into (A2) gives

$$\hat{\tau} = \frac{2d}{3s^4}(9s^4 + 2s^2 + 1), \qquad \hat{A} = \frac{2d}{5s^2}(3s^4 + 14s^2 + 3),$$
$$\hat{B} = \frac{4d}{s}(1 + s^2), \qquad \hat{K} = \frac{4d}{15s^4}(1 - s^2)(9s^4 + 6s^2 + 5).$$

For the normalized rate cv_r, define $A(z) := A(cv_r)$, with $z = (r\rho(r))^2$. The squared-rate integral is

$$A(z) = \frac{4\hat{\tau}\hat{A}}{\hat{B}^2} = \frac{(9z^2 + 2z + 1)(3z^2 + 14z + 3)}{15z^2(1 + z)^2},$$
$$A'(z) = -\frac{2(3z^2 - 2z + 3)(13z^2 + 6z + 1)}{15z^3(1 + z)^3} < 0.$$

The function $A(z)$ tends to ∞ as $z \downarrow 0$ and to 4 as $z \uparrow 1$. Thus every competing path with $A > 4$ has the same squared-rate integral as exactly one normalized rate cv_r. The comparison $K \le K(cv_r)$ then bounds $K/\sqrt{2A}$ by $K(cv_r)/\sqrt{2A(cv_r)}$. When $A = 4$, equality in Cauchy–Schwarz forces $a = b = 1/2$ almost everywhere, giving $K = 0$. This proves equality in (A5).

To evaluate the supremum over r in (A5), define the squared manipulation quotient of the rate v_r by

$$\begin{aligned} R(z) &:= \frac{K(v_r)^2}{A(v_r)B(v_r)}, \qquad z = (r\rho(r))^2, \\ &= \frac{\hat{K}^2}{\hat{\tau}^2\hat{A}\hat{B}}, \\ &= \frac{z^2(1 - z)(9z^2 + 6z + 5)^2}{10(1 + z)(9z^2 + 2z + 1)^2(3z^2 + 14z + 3)}. \end{aligned} \tag{A8}$$

The rational expression follows from (A2). Since $\hat{K} > 0$ for $0 < z < 1$, maximizing $R(z)$ also maximizes the quotient in (A5). Write $P(z) := 18z^3 + 39z^2 + 8z - 5$. Differentiating and factoring gives

$$R'(z) = -\frac{z(3z^2 - 2z + 3)(9z^2 + 6z + 5)(13z^2 + 6z + 1)P(z)}{5(1 + z)^2(9z^2 + 2z + 1)^3(3z^2 + 14z + 3)^2}.$$

Every factor other than $P(z)$ is positive on $(0, 1)$; in particular, $3z^2 - 2z + 3 = 3(z - 1/3)^2 + 8/3 > 0$. Thus $R'(z)$ has the opposite sign to $P(z)$. Since $P'(z) = 54z^2 + 78z + 8 > 0$, this polynomial is strictly increasing and vanishes at the root z_* defined in (53). Thus R increases on $(0, z_*)$ and decreases on $(z_*, 1)$, attaining its unique maximum at z_*. Reducing (A8) by (53) gives

$$R(z_*) = \frac{1584z_*^2 + 2046z_* - 535}{15360}.$$

The maximizing rate v_r has $(r\rho(r))^2 = z_*$, with $r \approx 0.33095$ and $\rho \approx 1.53767$. Restoring the factor $\lambda\sqrt{v_*}$ proves (54).

The reported decimals can be certified by rational arithmetic. Put

$$z_- = 0.25897201316507985454, \qquad z_+ = 0.25897201316507985455.$$

Exact evaluation gives $P(z_-) < 0 < P(z_+)$, hence $z_- < z_* < z_+$. The polynomial expression for $R(z_*)$ is increasing in $z_* > 0$. Evaluating it at these endpoints and comparing rational squares gives

$$0.081125738569 < \frac{\Xi_{1/2}}{\lambda\sqrt{v_*}} < 0.081125738570,$$

$$0.001645346364 < \frac{\Xi_{1/2}^2}{4\lambda^2 v_*} < 0.001645346365.$$

In particular, the normalized constant rounds to 0.08112574. The script certify_permanent_constant.py verifies these inequalities and the approximately 38-fold comparison using integer and rational arithmetic throughout. □

Appendix B. Certified kernel constants

We give error bounds for the numerical entries in Table 1. Throughout this appendix, $G(t) = t^{-\gamma}$, $0 < \gamma < 1$, and $\alpha := 1 - \gamma$. Let Q_n be the orthogonal projection onto functions constant on each of n equal cells, of width $h := 1/n$. In the orthonormal basis of normalized cell indicators, the matrix P_n representing $Q_n \mathcal{P} Q_n$ has entries

$$(P_n)_{ij} = \begin{cases} 0, & i < j, \\ F(h)/h, & i = j, \\ \big[F((d+1)h) - 2F(dh) + F((d-1)h)\big]/h, & d = i - j > 0, \end{cases} \qquad F(x) := \frac{x^{1+\alpha}}{\alpha(1+\alpha)}. \tag{A9}$$

These formulas integrate the singular diagonal cell exactly. The matrix $P_n + P_n^\top$ is symmetric and positive semidefinite, since it represents the restriction of the positive-type quadratic form to the cell-indicator space.

We treat both constants using a self-adjoint operator H: take $H = \mathcal{P} + \mathcal{P}^*$ for ν_G, and $H = i(\mathcal{P} - \mathcal{P}^*)$ on complex-valued $L^2([0,1])$ for a_G. In the latter case, complex conjugation pairs every eigenvalue with its negative. In either case, the constant of interest is half the largest eigenvalue of H. Write H_n for its cell matrix and λ_n for the largest eigenvalue of H_n.

Appendix B.1. Error from cell averaging

Lemma A1. *For either choice of H, put*

$$e_n := \frac{4n^{-\alpha}}{\alpha\sqrt{2\alpha+1}}.$$

Then $\|(I - Q_n)H\| \le e_n$. If q is the corresponding constant ν_G or a_G, and $q_n := \lambda_n/2$, then

$$q_n \le q \le q_n + \frac{e_n^2}{2(2q_n - e_n)}, \qquad 2q_n > e_n. \tag{A10}$$

Proof. Define $g(x) := x^{-\gamma}$ for $0 < x < 1$, and $g(x) := 0$ otherwise. Direct integration gives

$$\|g(\cdot + \delta) - g\|_{L^1(\mathbb{R})} = \frac{2\delta^\alpha}{\alpha}, \qquad 0 \le \delta \le 1.$$

For the two choices of H, use respectively the convolution kernels $k(x) := g(x) + g(-x)$ and $k(x) := i[g(x) - g(-x)]$. Extend $f \in L^2([0,1])$ by zero to $\mathbb{R}$, denoting this extension by $\widetilde{f}$, and put $u := k * \widetilde{f}$. Then $u|_{[0,1]} = Hf$, and Young's inequality gives

$$\|u(\cdot + \delta) - u\|_{L^2(\mathbb{R})} \le \frac{4\delta^\alpha}{\alpha}\|f\|_2.$$

Let D_δ be the set of t for which t and $t+\delta$ belong to the same grid cell. The identity for the variance about a cell average yields

$$\|(I-Q_n)Hf\|_2^2 = \frac{1}{h}\int_0^h \int_{D_\delta} |u(t+\delta)-u(t)|^2\,dt\,d\delta$$
$$\le \frac{16}{\alpha^2 h}\int_0^h \delta^{2\alpha}\,d\delta\,\|f\|_2^2 = e_n^2\|f\|_2^2.$$

Self-adjointness also gives $\|H(I-Q_n)\| \le e_n$. Thus $\|H - Q_nHQ_n\| \le 2e_n \to 0$, proving compactness.

For a unit vector write $f = x + y$, where $x = Q_nf$ and $y = (I-Q_n)f$. The preceding bounds imply

$$\langle f, Hf\rangle \le \lambda_n\|x\|_2^2 + 2e_n\|x\|_2\|y\|_2 + e_n\|y\|_2^2.$$

Consequently the largest eigenvalue of H is at most

$$\frac{\lambda_n + e_n + \sqrt{(\lambda_n - e_n)^2 + 4e_n^2}}{2} \le \lambda_n + \frac{e_n^2}{\lambda_n - e_n} \quad \text{when } \lambda_n > e_n.$$

The variational definition gives the lower bound λ_n. Dividing by two proves (A10). □

Appendix B.2. Verification of the finite matrices

The script `certify_kernel_constants.py` uses floating-point eigenvectors as candidates, then checks the resulting bounds with integer and rational arithmetic. We describe the checks so that both the numerical arithmetic and the choice of eigenvalue are verifiable. For the rational exponents in Table 1, integer root bounds enclose each $F(dh)/h$ between consecutive multiples of 2^{-80}. Taking the second differences in (A9) gives a rational Hermitian matrix A_n whose entrywise errors are at most 2^{1-80}. Hence

$$\|A_n - H_n\| \le \tau_n, \qquad \tau_n := 2n\,2^{-80}.$$

The skew case retains the exact form iK, with K real and skew-symmetric. Candidate eigenvectors have real and imaginary parts rounded to multiples of 2^{-40}.

First, a calculation on $m = 128$ cells bounds the second largest eigenvalue of H. Let $r_m > 0$ be an exactly evaluated Rayleigh quotient for A_m. Put $d_H := 1$ in the symmetric case and $d_H := 2$ in the skew case. A nonnegative number s satisfying

$$s^4 \ge \frac{\operatorname{tr}(A_m^4)}{d_H} - r_m^4$$

bounds the second largest eigenvalue of A_m. Indeed, the trace is the sum of fourth powers of all eigenvalues, and the largest positive eigenvalue is at least r_m; in the skew case the positive and negative eigenvalues occur in pairs. The second largest eigenvalue of H_m is therefore at most $s + \tau_m$. Restricting the preceding quadratic-form argument to vectors orthogonal to a largest-eigenvalue eigenvector of H_m shows that the second largest eigenvalue of H is at most any U satisfying

$$U > \max\{s+\tau_m, e_m\}, \qquad (U - s - \tau_m)(U - e_m) \ge e_m^2.$$

All these inequalities are checked with rational upper bounds for s and e_m.

For the finer matrix A_n, let $v \ne 0$ be its rational candidate eigenvector and compute exactly

$$r := \frac{v^*A_nv}{v^*v}, \qquad \delta^2 := \frac{v^*A_n^2v}{v^*v} - r^2.$$

Compression and the matrix error bound imply that the second largest eigenvalue of A_n is at most $U + \tau_n$. If $r > U + \tau_n$, its largest eigenvalue Λ_n satisfies

$$r \le \Lambda_n \le r + \frac{\delta^2}{r - U - \tau_n}.$$

To see the upper bound, expand v in an orthonormal eigenbasis. The expression $(\Lambda_n - x)(x - U - \tau_n)$ is nonpositive at every eigenvalue x; its average under the squared coefficients of $v/\|v\|_2$ gives $(\Lambda_n - r)(r - U - \tau_n) \le \delta^2$. Transferring back to H_n gives

$$\ell_n := r - \tau_n \le \lambda_n \le u_n := r + \frac{\delta^2}{r - U - \tau_n} + \tau_n. \tag{A11}$$

Thus the fully verified continuum enclosure is

$$\frac{\ell_n}{2} \le q \le \frac{u_n}{2} + \frac{e_n^2}{2(\ell_n - e_n)}, \qquad \ell_n > e_n.$$

The script checks each required strict inequality. Integer polynomial multiplication evaluates the Toeplitz matrix-vector products exactly; the fourth traces use direct integer matrix multiplication on the coarser grid. For each finite matrix, $u_n - \ell_n < 10^{-18}$. Outward rounding of the resulting continuum intervals gives

γ	n	interval for ν_G	interval for a_G
0.1	2048	$[0.58499122, 0.58499454]$	$[0.35511777, 0.35512324]$
0.25	2048	$[0.76336802, 0.76340853]$	$[0.42452126, 0.42459452]$
0.4	16384	$[1.04620433, 1.04624692]$	$[0.51760116, 0.51768780]$

Every number in each interval rounds to the corresponding entry of Table 1. The mass $c_G = 1/(1-\gamma)$ is evaluated analytically.

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